A Neural-Network-Assisted Approach to Recursive State Estimation for Energy Harvesting Complex Networks With Unknown Nonlinearities

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***Abstract*—This article investigates the problem of partial node-based (PNB) recursive state estimation for complex networks (CNs) with unknown nonlinearities and energy harvesting sensors. To mitigate the effects of energy constraints, an energy replenishment mechanism is employed, in which a group of energy harvesting sensors captures energy from the surrounding environment. These sensors transmit measurement outputs to remote state estimators only when their current energy levels are sufficient to cover the transmission energy costs. By exploiting the universal approximation property, neural networks (NNs) are utilized to approximate the unknown nonlinearities of the CNs. An NN-based recursive estimation algorithm is developed to simultaneously generate the estimates of the system state and the unknown nonlinearities. Following a specific set of recursions, the recursive state estimator gains and the NN weight (NNW) tuning parameters are calculated in a unified framework. Finally, the effectiveness of the developed recursive estimation algorithm is demonstrated through a simulation example.**

***Index Terms*—Energy harvesting mechanism (EHM), networked nonlinear systems, neural networks (NNs), recursive state estimation, unknown nonlinearities.**

Nomenclature

CNs	Complex networks.
PNB	Partial node-based.
SBC	Sector-bounded condition.
TE	Taylor expansion.
NN	Neural network.
EHM	Energy harvesting mechanism.
NNW	NN weight.
UB	Upper bound.
$\mathbb{R}^n$	n-dimensional Euclidean space.
$\mathbb{R}^{n\times m}$	Set of all $n \times m$ real matrices.
$\mathbb{N}$	Set of nonnegative integers.
$X \geq Y$	$X - Y$ is the positive semi-definite.
$X > Y$	$X - Y$ is the positive definite.
A^T	Transpose of the matrix A.
$\|A\|$	Frobenius norm of the matrix A.
$\mathrm{tr}\{A\}$	Trace of the matrix A.
$\mathbb{P}\{\cdot\}$	Occurrence probability of the random event "$\cdot$."
$\mathbb{E}\{x\}$	Expectation of the stochastic variable x.
$\mathrm{diag}\{\cdots\}$	Block-diagonal matrix.
$Z \otimes Q$	Kronecker product of Z and Q.
$\mathbb{E}\{x\|y\}$	Expectation of x conditional on y.

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I. Introduction

A TYPICAL CN is composed of numerous spatially distributed nodes that are interconnected through specific types of topologies, whereby information exchange is achieved among the nodes via these connections [7], [44], [49]. The unique network architecture and dynamic behavior of CNs make them particularly effective for modeling various dynamical systems, including social networks, power grids, the World Wide Web, and biological networks [45]. It should be noted that state estimation represents a fundamental aspect of CN analysis, as the acquisition of state information serves as the prerequisite for conducting practical tasks such as optimal control, tracking control, and consensus [5], [38], [43], [51], [52], [58]. In practical engineering, however, full access to the state information of a CN is often unavailable; only the measurement signals from sensor nodes can be obtained, which significantly complicates the state estimation process in CNs [14], [15], [33], [55]. As a result, the development of effective state estimation strategies is of considerable theoretical and practical significance to enhance the performance and reliability of CNs [17], [29], [37].

An implicit assumption in most existing CN state estimation studies is that the measurement outputs from all sensor nodes are available [11]. This assumption, however, is highly inconsistent with actual engineering practice due to various factors, including unreliable communications, inaccurate system models, harsh operating environments, and the large scale of CNs [12]. To overcome these challenges, a PNB state estimation strategy has been proposed, whereby state estimation tasks are carried out using the measurement signals collected from partial nodes only [26], [60]. It is worth noting that the PNB state estimation strategy contributes to reduced budgets in comparison with the full node-based estimation approaches [54]. Accordingly, increasing attention has been directed toward utilizing the PNB state estimation strategy to address the estimation problems of CNs. For example, the PNB state estimation strategy has been investigated for the CNs with gain variations in [20]. In [27], the PNB state estimation issue has been analyzed for stochastic CNs with switched topology. By utilizing the information fusion approach, the PNB state estimation problem for time-delayed stochastic CNs has been addressed in [30].

The nonlinear estimation problem has emerged as a significant focus within the systems and control research communities [16], [39], [56]. Numerous methods have been proposed to address the challenges posed by the underlying nonlinear dynamics [1], [2], [3], [4], [21], [23], [53]. These methods are generally categorized into three main groups, namely, the SBC-based methods [31], the TE-based approaches [35], and the NN-based techniques [34], [59]. In the SBC-based methods, the nonlinearities are addressed by employing sector-bounded-like conditions, which are essentially inequalities. The TE-based approaches perform the linearization of nonlinear functions by using Taylor polynomials. Both approaches, however, rely heavily on information regarding the nonlinearities, which is often difficult to obtain in practical applications due to harsh external environments, complex internal networks, and uncontrollable uncertainty [19]. Consequently, the NN-based techniques are widely adopted due to their capability to approximate unknown nonlinearities [47], [57]. This advantage is particularly notable when the information on nonlinear dynamics is unavailable. Despite their theoretical potential, state estimation problems employing NNs in CNs with partially accessible nodes and unknown nonlinear dynamics remain insufficiently explored, highlighting a critical area for further investigation.

Energy constraints constitute an inherently significant issue in the communication process [8], [36]. In engineering practice, the electric energy required by communication devices is supplied by batteries with limited power capacity [41]. For CNs or sensor networks, however, powering each sensor with such limited-capacity batteries is both technically impractical and economically unsustainable [10]. To mitigate the impacts of energy constraints, an EHM is adopted [24]. The EHM employs a set of energy harvesting sensors that collect energy from the external environment and store it for the purpose of measurement transmission [6]. These sensors transmit measurement signals to remote state estimators only when their current energy levels are sufficient to cover the transmission energy costs [48]. Moreover, the energy obtained by the energy harvesting sensors is typically stochastic, owing to the unpredictability of energy sources and complex environmental disturbances. This energy uncertainty may result in intermittent transmission issues, which, if not properly addressed, could significantly degrade the estimation performance of the estimator.

Despite the theoretical and practical significance of NN-based state estimation, limited research has been conducted on the use of NNs for solving recursive state estimation problems in CNs, particularly considering the simultaneous challenges of partially accessible nodes and EHMs. The primary objective of this article is to address this research gap. Based on the preceding discussions, the focus of this study is directed toward the PNB state estimation problem for CNs with unknown nonlinearities and EHMs. Addressing this estimator design issue involves three practical or theoretical challenges: 1) how the influences of unreliable transmission behaviors can be mitigated to improve the estimation performance; 2) how an appropriate law can be formulated to update the NNWs for the unknown nonlinearities widely existing in CNs with partially accessible nodes; and 3) how an easy-to-implement NN-based estimation algorithm with both theoretically and practical meaning can be developed for CNs with unknown nonlinearities and EHMs.

The primary contributions of this article are summarized from the following three aspects.

1) The NN-based recursive state estimation problem is addressed, for the first time, in the context of a class of nonlinear CNs with partially accessible nodes and EHMs, where both the unknown nonlinearities and energy constraints are considered simultaneously to enhance estimation robustness under realistic conditions.
2) An NN approximation method and a covariance-based NNW updating approach are proposed to effectively deal with the unknown nonlinear dynamics present in the CNs, which ensures that the nonlinearities can be accurately approximated without requiring prior knowledge or explicit modeling of the system.
3) An easy-to-implement algorithm is developed for recursively calculating the desired estimator gains and the NNW tuning scalars in a unified framework, which enables online adaptation of both the estimation process and the NNW adjustments, thereby improving computational efficiency and practical applicability.

The remainder of this article is organized as follows. In Section II, the CN, the EHM, the proposed NN-based recursive estimation strategy, and the design objective are formulated. Section III presents four theorems that are established to guarantee, simultaneously, the minimization of the UBs for the covariance of the estimation errors of both the NNWs and the system states. In Section IV, a numerical example is provided to verify the efficacy of the proposed estimation strategy. Finally, the conclusion is presented in Section V.

II. Problem Formulation and Preliminaries

A. Nonlinear System Model

In this article, we consider the following class of discrete-time CNs consisting of S coupled nodes:

$$x_{i,k+1} = A_i x_{i,k} + \sum_{i=1}^{S} \pi_{i,j} \Lambda x_{j,k} + f\left(x_{i,k}\right) + B_i \omega_{i,k} \tag{1}$$

where $x_{i,k} \in \mathbb{R}^{n_x}$ denotes the state vector of the ith node, $i \in \{1, 2, \ldots, S\}$. $\Theta \triangleq [\pi_{ij}]_{S \times S} \in \mathbb{R}^{S \times S}$ is the weight adjacency matrix of the CN with $\pi_{ij} > 0 (i \neq j)$ if nodes i is capable of receiving signals from node j. $\Lambda \triangleq \mathrm{diag}\{\gamma_1, \gamma_2, \ldots, \gamma_n\} \geq 0$ represents the inner coupling matrix that connects the jth node. Here, the graph under consideration is directed. $f_i(\cdot) \in \mathbb{R}^{n_f}$ is an unknown nonlinear function defined on a compact set $\Omega \in \mathbb{R}^n$. It is assumed that $\|f_i(\cdot)\| \leq \bar{f}_i$, where $\bar{f}_i$ is a known constant. A_i, B_i, and E_i are known constant matrices. $\omega_{i,k} \in \mathbb{R}^{n_\omega}$ denotes the process noise.

This article aims to estimate the states of the CN (1) using the available measurement outputs. Due to factors such as the extremely high number of nodes, elevated access costs, and unreliable communication conditions, it is impractical to obtain the measurement outputs of all nodes in a CN [20]. Therefore, a practical approach is to estimate the states of the CN by utilizing the measurement signals obtained from only partially accessible nodes.

Without loss of generality, it is assumed that access is available to the outputs of the first s_0 nodes. For $i \in \{1, 2, \ldots, s_0\}$, the following holds:

$$y_{i,k} = C_i x_{i,k} + D_i \upsilon_{i,k} \tag{2}$$

where $y_{i,k} \in \mathbb{R}^{n_y}$ denotes the output of the ith node, $C_i \in \mathbb{R}^{n_y \times n_x}$ and $D_i \in \mathbb{R}^{n_y \times q}$ are known constant matrices, and $\upsilon_{i,k} \in \mathbb{R}^{n_\upsilon}$ represents the measurement noise.

The disturbance noises $\omega_{i,k} \in \mathbb{R}^{n_\omega}$ and $\upsilon_{i,k} \in \mathbb{R}^{n_\upsilon}$ are random, zero-mean, and have covariances $Q_{i,k}$ and $G_{i,k}$, respectively. It is assumed that $\omega_{i,k}$ and $\upsilon_{i,k}$ are mutually uncorrelated and satisfy the following statistical properties:

$$\|\omega_{i,k}\| \leq \bar{\omega}_i, \quad \|\upsilon_{i,k}\| \leq \bar{\upsilon}_i$$

where $\bar{\omega}$ and $\bar{\upsilon}$ are constants that are independent of the initial state.

B. Energy Harvesting Sensors

In this article, communication between the energy harvesting sensors and the recursive state estimator is shown in Fig. 1. Each sensor harvests energy from its surroundings and stores it in the battery. When the stored unit energy is nonzero, the sensor transmits the measurement outputs to the recursive state estimator while consuming one unit of energy.

For the ith node, the energy level of sensor i at time instant k is denoted as $\aleph_{i,k} \in \{0, 1, 2, \ldots, H_i\}$, where H_i indicates the maximum number of energy units that sensor i can store. The sequence $\{h_{i,k}\}_{k \geq 0}$ represents the energy harvested by sensor i, which is assumed to be known at each time instant. It is further assumed that $\{h_{i,k}\}_{k \geq 0}$ forms a sequence of independent

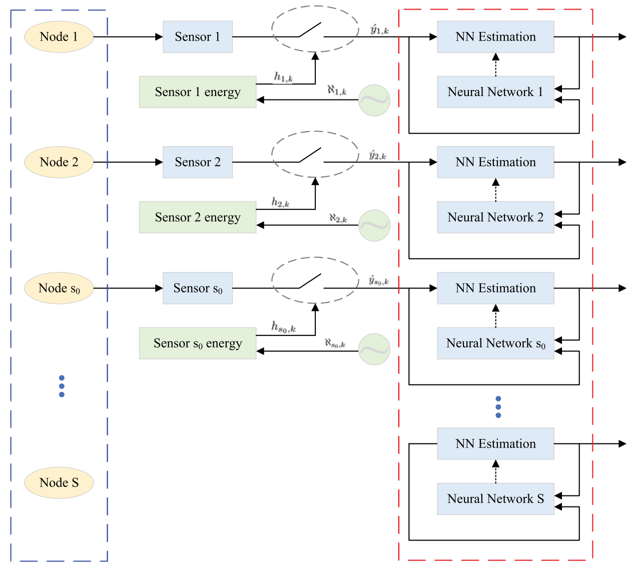

Fig. 1. NN-based recursive state estimation with energy harvesting sensors.

and identically distributed random variables. The probability distribution of $h_{i,k}$ is given by the following equation:

$$\mathbb{P}\left\{h_{i,k} = l\right\} = \mu_l, \quad l = 0, 1, 2 \cdots \tag{3}$$

where μ_l satisfies $0 \leq \mu_l \leq 1$ and $\sum_{l=0}^{+\infty} \mu_l = 1$. It is obvious that $0 \leq h_{i,k} \leq H_i$.

Define an indicator variable $\Re_{i,\aleph_{i,k}}$ as follows:

$$\Re_{i,\aleph_{i,k}} \triangleq \begin{cases} 1, & \text{if } \aleph_{i,k} > 0, i \in \{1, 2, \ldots, s_0\} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Here, $\Re_{i,\aleph_{i,k}}$ acts as an indicator function. When $\Re_{i,\aleph_{i,k}} = 1$, the signal transmission is considered successful; when $\Re_{i,\aleph_{i,k}} = 0$, the signal transmission is deemed to have failed. The dynamics of $\aleph_{i,k}$ can be written as follows:

$$\begin{cases} \aleph_{i,k+1} = & \max\left\{\min\left\{\aleph_{i,k} + h_{i,k} - \Re_{i,\aleph_{i,k}}, H_i\right\}, 0\right\} \\ \aleph_{i,0} \leq & H_i. \end{cases} \tag{5}$$

The measurement received by the remote state estimator is defined as $\acute{y}_{i,k}$, which is expressed by the following equation:

$$\acute{y}_{i,k} = \Re_{i,\aleph_{i,k}} y_{i,k}. \tag{6}$$

C. NN-Based Recursive State Estimator

By considering the universal approximation property [19], an NN is employed to approximate the unknown nonlinear function $f_i(\cdot)$ with a sufficiently small approximation error such that $f(x_{i,k}) = W_i \zeta_i(x_{i,k}) + \rho_{i,k}$. Here, $W_i \in \mathbb{R}^{n_x \times n_x}$, $\zeta(\cdot) \in \mathbb{R}^{n_x}$, and $\rho_{i,k} \in \mathbb{R}^n$ denote the ideal weight matrix, the activation function, and the approximation error of the NN, respectively. Thus, the system can be rewritten as follows:

$$\begin{cases} x_{i,k+1} = A_i x_{i,k} + \sum_{i=1}^{S} \pi_{i,j} \Lambda x_{j,k} + W_i \zeta_i\left(x_{i,k}\right) \\ +B_i \omega_{i,k} + \rho_{i,k}, \quad i \in \{1, 2, \ldots, S\} \\ y_{i,k} = C_i x_{i,k} + D_i \upsilon_{i,k}, \quad i \in \{1, 2, \ldots, s_0\}. \end{cases} \tag{7}$$

Assumption 1: The connecting NNW matrix W_i, the NN activation function $\zeta_i(\cdot)$, and the NN approximate error $\rho_{i,k}$ satisfy the following conditions:

$$\|W_i\| \leq \bar{W}_i, \quad \|\zeta_i(\cdot)\| \leq \bar{\zeta}_i, \quad \|\rho_{i,k}\| \leq \bar{\rho}_i$$

where $\bar{W}_i$, $\bar{\zeta}_i$, and $\bar{\rho}_i$ are known positive constants.

Remark 1: The unknown nonlinearities considered in this article are assumed to be bounded. It has been demonstrated in [9] that NNs are capable of approximating the unknown nonlinearities with arbitrary precision when the nonlinear functions are continuous. Therefore, the assumption that the connecting matrix W_i, the activation function $\zeta_i(\cdot)$, and the approximation error $\rho_{i,k}$ are bounded is reasonable.

Let $\hat{x}_{i,k}$ denote the estimate of $x_{i,k}$ and $\hat{W}_{i,k}$ denote the estimate of W_i at time instant k. The estimator gain to be designed is denoted as $L_{i,k}$. Based on the above considerations, the following NN-based recursive estimator is proposed to address the PNB state estimation problem for the CNs with EHM and unknown nonlinearities:

$$\begin{cases} \hat{x}_{i,k+1} = A_i\hat{x}_{i,k} + \sum_{i=1}^{S}\pi_{i,j}\Lambda\hat{x}_{j,k} + \hat{W}_{i,k}\zeta_i(\hat{x}_{i,k}) + L_{i,k} \\ \times\left(\acute{y}_{i,k} - \varpi_{i,k}C_i\hat{x}_{i,k}\right), \ i \in \{1,2,\ldots,s_0\} \\ \hat{x}_{i,k+1} = A_i\hat{x}_{i,k} + \sum_{i=1}^{S}\pi_{i,j}\Lambda\hat{x}_{j,k} + \hat{W}_{i,k}\zeta_i(\hat{x}_{i,k}) \\ \quad i \in \{s_0+1,\ldots,S\}. \end{cases} \tag{8}$$

The following adaptive tuning law of the NNW is defined as:

$$\begin{cases} \hat{W}_{i,k} = \hat{W}_{i,k-1} - \theta_{i,k}C_i^T\left(\acute{y}_{i,k} - \varpi_{i,k}C_i\hat{x}_{i,k}\right) \\ \times\tilde{\zeta}_i^T(\hat{x}_{i,k-1}), \ i \in \{1,2,\ldots,s_0\} \\ \hat{W}_{i,k} = \hat{W}_{i,k-1}, \ i \in \{s_0+1,\ldots,S\} \end{cases} \tag{9}$$

where $\theta_{i,k}$ denotes the tuning parameter to be designed and $\tilde{\zeta}_i(\hat{x}_{i,k-1}) = \zeta_i(\hat{x}_{i,k-1})/(\|1 + \zeta_i^T(\hat{x}_{i,k-1})\zeta_i(\hat{x}_{i,k-1})\|\|C_i^TC_i\|)$. The term $\varpi_{i,k} \triangleq \mathbb{E}\{\Re_{i,\aleph_{i,k}}\}$ represents the expected value of the indicator function, the detailed characteristics of which will be introduced in Section III-A.

Remark 2: The proposed NN-based recursive estimator with recursively calculated gain parameters $L_{i,k}$ and NNW tuning parameters $\theta_{i,k}$ possesses significant advantages in enhancing adaptability from the perspective of practical engineering, which distinguishes it from existing state estimation strategies. These advantages are reflected in the following three aspects: 1) the developed approach integrates the NN method and the recursive estimator approach, whereby the gain parameters and the NNW tuning parameters of the estimator are calculated recursively within a unified framework; 2) in contrast to traditional NN approximation approaches that adopt fixed gain parameters, the NN tuning parameters in this article are computed recursively, which contributes to improved approximation performance; 3) the NN-based state estimation scheme is established for systems with unknown nonlinear dynamics while maintaining applicability to linear systems and systems with known nonlinear dynamics; and 4) under specific scenes such as sensor or actuator failures, the proposed NN-based recursive estimator also contributes to satisfactory approximation performance, which gives the credit to the recursively calculated gain parameters and the NNW tuning parameters.

Denote

$$\begin{aligned}
x_k &\triangleq \left[x_{1,k}^T \ x_{2,k}^T \ \cdots \ x_{S,k}^T\right]^T \in \mathbb{R}^{Sn_x} \\
\hat{x}_k &\triangleq \left[\hat{x}_{1,k}^T \ \hat{x}_{2,k}^T \ \cdots \ \hat{x}_{S,k}^T\right]^T \in \mathbb{R}^{Sn_x} \\
\omega_k &\triangleq \left[\omega_{1,k}^T \ \omega_{2,k}^T \ \cdots \ \omega_{S,k}^T\right]^T \in \mathbb{R}^{Sn_\omega} \\
\rho_k &\triangleq \left[\rho_{1,k}^T \ \rho_{2,k}^T \ \cdots \ \rho_{S,k}^T\right]^T \in \mathbb{R}^{Sn_x} \\
W &\triangleq \operatorname{diag}\{W_1, W_2, \ldots, W_S\} \in \mathbb{R}^{Sn_x \times Sn_x} \\
A &\triangleq \operatorname{diag}\{A_1, A_2, \ldots, A_S\} \in \mathbb{R}^{Sn_x \times Sn_x} \\
B &\triangleq \operatorname{diag}\{B_1, B_2, \ldots, B_S\} \in \mathbb{R}^{Sn_x \times Sn_\omega} \\
\bar{C} &\triangleq \operatorname{diag}\left\{C_1, C_2, \ldots, C_{s_0}\right\} \in \mathbb{R}^{s_0n_y \times s_0n_x} \\
\bar{D} &\triangleq \operatorname{diag}\left\{D_1, D_2, \ldots, D_{s_0}\right\} \in \mathbb{R}^{s_0n_\upsilon \times s_0n_\upsilon} \\
\bar{\varpi}_k &\triangleq \operatorname{diag}\left\{\varpi_{1,k}, \varpi_{2,k}, \ldots, \varpi_{s_0,k}\right\} \in \mathbb{R}^{s_0n_y \times s_0n_y} \\
\hat{W}_k &\triangleq \operatorname{diag}\left\{\hat{W}_{1,k}, \hat{W}_{2,k}, \ldots, \hat{W}_{S,k}\right\} \in \mathbb{R}^{Sn_x \times Sn_x} \\
\bar{\upsilon}_k &\triangleq \left[\upsilon_{1,k}^T \ \upsilon_{2,k}^T \ \cdots \ \upsilon_{l_0,k}^T\right]^T \in \mathbb{R}^{s_0n_\upsilon} \\
\bar{y}_k &\triangleq \left[y_{1,k}^T \ y_{2,k}^T \ \cdots \ y_{s_0,k}^T\right]^T \in \mathbb{R}^{s_0n_y} \\
\acute{y}_k &\triangleq \left[\acute{y}_{1,k}^T \ \acute{y}_{2,k}^T \ \cdots \ \acute{y}_{s_0,k}^T\right]^T \in \mathbb{R}^{s_0n_y} \\
\acute{\Re}_k &\triangleq \left[\Re_{1,\aleph_{1,k}} \ \Re_{2,\aleph_{2,k}} \ \cdots \ \Re_{s_0,\aleph_{s_0,k}}\right]^T \in \mathbb{R}^{s_0n_y} \\
\check{y}_k &\triangleq \left[\acute{y}_k^T \ 0\right]^T \in \mathbb{R}^{Sn_y}, y_k \triangleq \left[\bar{y}_k^T \ 0\right]^T \in \mathbb{R}^{Sn_y} \\
\Re_k &\triangleq \left[\acute{\Re}_k^T \ 0\right]^T \in \mathbb{R}^{Sn_y}, \check{C} \triangleq \operatorname{diag}\left\{\bar{C}, 0\right\}, \in \mathbb{R}^{Sn_y \times Sn_x} \\
\upsilon_k &\triangleq \left[\bar{\upsilon}_k^T \ 0\right]^T \in \mathbb{R}^{Sn_\upsilon}, \check{D} \triangleq \operatorname{diag}\left\{\bar{D}, 0\right\}, \in \mathbb{R}^{Sn_y \times Sn_\upsilon} \\
\varpi_k &\triangleq \operatorname{diag}\{\bar{\varpi}_k, 0\}, \in \mathbb{R}^{Sn_y \times Sn_y} \\
\mathcal{F}(x_k) &\triangleq \left[f^T(x_{1,k}) \ f^T(x_{2,k}) \ \cdots \ f^T(x_{S,k})\right]^T \in \mathbb{R}^{Sn_f} \\
\zeta(x_k) &\triangleq \left[\zeta_1^T(x_{1,k}) \ \zeta_2^T(x_{2,k}) \ \cdots \ \zeta_S^T(x_{S,k})\right]^T \in \mathbb{R}^{Sn_x} \\
\zeta(\hat{x}_k) &\triangleq \left[\zeta_1^T(\hat{x}_{1,k}) \ \zeta_2^T(\hat{x}_{2,k}) \ \cdots \ \zeta_S^T(\hat{x}_{S,k})\right]^T \in \mathbb{R}^{Sn_x} \\
\tilde{\zeta}(\hat{x}_k) &\triangleq \left[\tilde{\zeta}_1^T(\hat{x}_{1,k}) \ \tilde{\zeta}_2^T(\hat{x}_{2,k}) \ \cdots \ \tilde{\zeta}_S^T(\hat{x}_{S,k})\right]^T \in \mathbb{R}^{Sn_x}.
\end{aligned}$$

Then, we have

$$\begin{cases} x_{k+1} = \check{A}x_k + \mathcal{F}(x_k) + B\omega_k \\ y_k = \check{C}x_k + \check{D}\upsilon_k \end{cases} \tag{10}$$

where $\check{A} \triangleq A + (\Theta \otimes \Lambda)$. Moreover, by denoting

$$\begin{aligned} Q_k &\triangleq \operatorname{diag}\left\{Q_{1,k}, Q_{2,k}, \ldots, Q_{S,k}\right\} \\ \bar{G}_k &\triangleq \operatorname{diag}\left\{G_{1,k}, G_{1,k}, \ldots, G_{s_0,k}\right\}, \ G_k \triangleq \operatorname{diag}\left\{\bar{G}_k, 0\right\} \end{aligned}$$

we obtain

$$\mathbb{E}\left\{\omega_k\omega_k^T\right\} = Q_k, \ \mathbb{E}\left\{\upsilon_k\upsilon_k^T\right\} = G_k.$$

In this case, the NN-based recursive estimator can be represented as follows:

$$\begin{cases} \hat{x}_{k+1} = \check{A}\hat{x}_k + \hat{W}_k\zeta(\hat{x}_k) + L_k\left(\check{y}_k - \check{C}\hat{x}_k\right) \\ \hat{W}_k = \hat{W}_{k-1} + \theta_k\check{C}^T\left(\check{y}_k - \check{C}\hat{x}_k\right)\tilde{\zeta}^T(\hat{x}_{k-1}). \end{cases} \tag{11}$$

Next, letting the estimation error of the NNW be $\tilde{W}_k \triangleq W - \hat{W}_k$, we have

$$\begin{aligned} &\tilde{W}_{k+1} \\ &= \tilde{W}_k + \theta_{k+1}\check{C}^T\varpi_{k+1}\check{C}L_k\left(\Re_k - \varpi_k\right)\check{C}x_k\tilde{\zeta}^T(\hat{x}_k) \end{aligned}$$

$$
\begin{aligned}
&-\theta_{k+1}\check{C}^T\left(\Re_{k+1}-\varpi_{k+1}\right)\check{C}\\
&\times\left(W\zeta(x_k)+\check{A}x_k+B\omega_k+\check{C}\rho_k\right)\tilde{\zeta}^T(\hat{x}_k)\\
&-\theta_{k+1}\check{C}^T\varpi_{k+1}\check{C}\left(\check{A}\tilde{x}_k+\tilde{W}_k+B\omega_k\right.\\
&\qquad\left.+\varepsilon_k-L_k\varpi_k\check{C}\tilde{x}_k-L_k\Re_k\check{D}\upsilon_k\right)\tilde{\zeta}^T(\hat{x}_k)\\
&-\theta_{k+1}\check{C}^T\times\Re_{k+1}\check{D}\upsilon_{k+1}\tilde{\zeta}^T(\hat{x}_k)
\end{aligned}\tag{12}
$$

where $\varepsilon_k \triangleq W(\zeta(x_k)-\zeta(\hat{x}_k))+\rho_k$.

Defining $\tilde{x}_k \triangleq x_k-\hat{x}_k$ as the state estimation error, the error dynamics is governed by the following equation:

$$
\begin{aligned}
\tilde{x}_{k+1}=\check{A}\tilde{x}_k+\tilde{W}_k\zeta(\hat{x}_k)+B\omega_k+\varepsilon_k-L_k\left(\varpi_k\check{C}\tilde{x}_k\right.\\
\left.+\left(\Re_k-\varpi_k\right)\check{C}x_k+\Re_k\check{D}\upsilon_k\right).
\end{aligned}\tag{13}
$$

Moreover, by denoting

$$
\begin{aligned}
&Q_k \triangleq \operatorname{diag}\left\{Q_{1,k},Q_{2,k},\ldots,Q_{S,k}\right\}\\
&\bar{G}_k \triangleq \operatorname{diag}\left\{G_{1,k},G_{1,k},\ldots,G_{s_0,k}\right\},G_k \triangleq \operatorname{diag}\left\{\bar{G}_k,0\right\}
\end{aligned}
$$

one has

$$
\mathbb{E}\left\{\omega_k\omega_k^T\right\}=Q_k,\ \mathbb{E}\left\{\upsilon_k\upsilon_k^T\right\}=G_k.
$$

Now, we are ready to highlight the two main objectives of this article as follows.

1) To design the NNW adaptive tuning law for a class of nonlinear CNs with partially accessible nodes and EHMs, aiming to achieve enhanced approximation performance of the unknown nonlinearities.
2) To develop an effective algorithm for recursively calculating the estimator gain L_k and the NNW tuning parameter θ_k within a unified framework such that the UBs of $P_k \triangleq \mathbb{E}\{\tilde{x}_k\tilde{x}_k^T\}$ and $\Phi_k \triangleq \operatorname{tr}\{\mathbb{E}\{\tilde{W}_k\tilde{W}_k^T\}\}$ are locally minimized at each time instant.

III. Main Results

A. Design of the NN Tuning Scalar

Before proceeding further, the following lemmas are introduced, which will be utilized in the subsequent estimator design process.

Lemma 1 ([25]): For any real-valued matrices $\mathscr{G}_1$ and $\mathscr{G}_2$ with any scalar $g>0$, the following inequality holds:

$$
\mathscr{G}_1\mathscr{G}_2^T+\mathscr{G}_2\mathscr{G}_1^T \leq g\mathscr{G}_1\mathscr{G}_1^T+g^{-1}\mathscr{G}_2\mathscr{G}_2^T. \tag{14}
$$

Lemma 2 ([42]): For any $k>0$, assume that $M=M^T>0$, $G_k(M)=G_k^T(M)\in\mathbb{R}^{n\times n}$, and $\mathcal{Z}_k(M)=\mathcal{Z}_k^T(M)\in\mathbb{R}^{n\times n}$. If $G_k(J)\geq G_k(M)$, for any $M\leq J=J^T$ and $\mathcal{Z}_k(J)\geq G_k(J)$, the solutions $\mathcal{V}_k$ and $\mathcal{Q}_k$ to the following difference equations:

$$
\mathcal{V}_{k+1}=G_k(\mathcal{V}_k),\ \mathcal{Q}_{k+1}=\mathcal{Z}_k(\mathcal{Q}_k),\ \mathcal{V}_0=\mathcal{Q}_0
$$

satisfy $\mathcal{V}_k\leq\mathcal{Q}_k$.

Lemma 3: Let $\aleph_{i,k}$ be the energy level whose distribution follows (5). Define

$$
\phi_{i,k} \triangleq \left[\mathbb{P}\left\{\aleph_{i,k}=0\right\}\ \mathbb{P}\left\{\aleph_{i,k}=1\right\}\ \cdots\ \mathbb{P}\left\{\aleph_{i,k}=H_i\right\}\right]^T
$$

and the recursion of the probability distribution $\phi_{i,k}$ is given by the following equation:

$$
\phi_{i,k+1}=\tau_i+\Psi_i\phi_{i,k}
$$

with the initial condition

$$
\phi_{i,0} \triangleq \begin{bmatrix}\underbrace{0\ \cdots\ 0}_{\aleph_{i,0}} & 1 & \underbrace{0\ \cdots\ 0}_{H_i-\aleph_{i,0}}\end{bmatrix}^T
$$

where

$$
\tau_i \triangleq \begin{bmatrix}\underbrace{0\ \cdots\ 0}_{H_i} & 1\end{bmatrix}^T
$$

$$
\Psi_i \triangleq -\begin{bmatrix}
-\mu_0 & -\mu_0 & 0 & \cdots & 0\\
-\mu_1 & -\mu_1 & -\mu_0 & \cdots & 0\\
-\mu_2 & -\mu_2 & -\mu_1 & \cdots & 0\\
\vdots & \vdots & \vdots & \ddots & \vdots\\
-\mu_{H_i-1} & -\mu_{H_i-1} & -\mu_{H_i-2} & \cdots & -\mu_0\\
\sum_{z=0}^{H_i-1}\mu_j & \sum_{z=0}^{H_i-1}\mu_j & \sum_{z=0}^{H_i-2}\mu_j & \cdots & \mu_0
\end{bmatrix}.
$$

It is easy to obtain from Lemma 3 that

$$
\varpi_{i,k}=\mathbb{P}\left\{\Re_{i,\aleph_{i,k}}=1\right\}=\begin{bmatrix}0 & \underbrace{1\ \cdots\ 1}_{H_i}\end{bmatrix}\phi_{i,k}. \tag{15}
$$

Theorem 1: Let positive scalars β_1–β_4 be given. Assume that there exists a set of real-valued matrices $\bar{\Phi}_k$ with the initial condition $\bar{\Phi}_0=\Phi_0$ satisfying the recursive equation (16), as shown at the bottom of the next page.

Then, $\bar{\Phi}_k$ is a UB of the trace of the NNW estimation error covariance Φ_k.

Proof: Denote the trace of the estimation error covariance of NNW as follows:

$$
\Phi_k \triangleq \operatorname{tr}\left\{\mathbb{E}\left\{\tilde{W}_k\tilde{W}_k^T\right\}\right\}.
$$

Then, Φ_{k+1} can be calculated in (17), shown at the bottom of page 7.

With the help of Lemma 1, we obtain the inequality (18), as shown at the bottom of page 8, from (17) by using some positive scalars $\{\beta_i\}_{i=1,2,3,4}$ (18), where $\bar{\mathcal{F}}=S\bar{f}^2$. Finally, it follows from (16) that:

$$
\Phi_{k+1}\leq\bar{\Phi}_{k+1}
$$

which ends the proof. ■

Now, the design of the NNW tuning parameter can be proceeded with in a recursive manner, ensuring that the UB derived in Theorem 1 is minimized.

Theorem 2: The UB of the trace of NNW estimation error covariance is minimized with the NNW tuning parameter (19), as shown at the bottom of page 9.

Proof: Taking the partial derivative of $\bar{\Phi}_{k+1}$ with respect to θ_{k+1} and letting

$$
\frac{\partial\bar{\Phi}_{k+1}}{\partial\theta_{k+1}}=0
$$

we have the results as shown in the equation at the bottom of page 10.

Then, it is easy to see that $\bar{\Phi}_{k+1}$ is minimized if the NN tuning scalar θ_{k+1} is selected as (19).

Remark 3: At this stage, the design of the NNW tuning parameter θ_k for the unknown nonlinearities has been completed. The parameter θ_k is computed recursively by minimizing the trace of the estimation error covariance of the NNW, denoted as Φ_k. In contrast to the time-invariant NNW tuning parameter, the recursively computed parameter possesses the capability to achieve improved approximation performance, as the unknown nonlinearities exhibit transient properties. Consequently, the minimization of Φ_k is justified from both theoretical and practical perspectives. ■

B. Design of the Estimator Gain Matrix

In this section, an effective algorithm is developed to parameterize the NN-based recursive estimator gains, aiming to minimize the UB of the estimation error covariance of the system states.

Theorem 3: Let positive scalars β_5 and β_6 be given. Assume that there exists a set of real-valued matrices $\bar{P}_k$ (with the initial condition $\bar{P}_0 = P_0$) satisfying the following recursive equation:

$$\begin{aligned}\bar{P}_{k+1} &= (1+\beta_5+\beta_6)\left(\check{A}-L_k\varpi_k\check{C}\right)\bar{P}_k\left(\check{A}^T-\check{C}^T\varpi_k^T L_k^T\right)\\&\quad+\left(1+\beta_5^{-1}+\beta_5\right)\bar{\mathcal{F}}I+\left(1+\beta_5^{-1}\right)^2\hat{W}_k\zeta(\hat{x}_k)\zeta^T(\hat{x}_k)\hat{W}_k^T\\&\quad+\left(2+\beta_6^{-1}+\beta_6\right)L_k\left(\Re_k-\varpi_k\right)\check{C}\bar{P}_k\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T\\&\quad+\left(1+\beta_6^{-1}\right)^2L_k\left(\Re_k-\varpi_k\right)\check{C}\hat{x}_k\hat{x}_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T\\&\quad+BQ_kB^T+L_k\Re_k\check{D}G_k\check{D}^T\Re_k^TL_k^T. \qquad (20)\end{aligned}$$

Then, $\bar{P}_k$ is a UB of the estimation error covariance P_k.

Proof: The estimation error covariance P_{k+1} is calculated as follows:

$$\begin{aligned}P_{k+1} &= \mathbb{E}\left\{\tilde{x}_{k+1}\tilde{x}_{k+1}^T\right\}\\&= \mathbb{E}\Big\{\left(\check{A}-L_k\varpi_k\check{C}\right)P_k\left(\check{A}^T-\check{C}^T\varpi_k^TL_k^T\right)\\&\qquad+2\left(\check{A}-L_k\varpi_k\check{C}\right)\tilde{x}_k\left(\mathcal{F}^T(x_k)-\zeta^T(\hat{x}_k)\hat{W}_k^T\right)\\&\qquad+2\left(\check{A}-L_k\varpi_k\check{C}\right)\tilde{x}_kx_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T\\&\qquad+B\omega_k\omega_k^TB^T+L_k\Re_k\check{D}\upsilon_k\upsilon_k^T\check{D}^T\Re_k^TL_k^T\Big\}. \qquad (21)\end{aligned}$$

With the aid of Lemma 1, it is obtained from (21) that

$$\begin{aligned}&P_{k+1}\\&\le \mathbb{E}\Big\{(1+\beta_5+\beta_6)\left(\check{A}-L_k\varpi_k\check{C}\right)\tilde{x}_k\tilde{x}_k^T\left(\check{A}^T-\check{C}^T\varpi_k^TL_k^T\right)\\&\qquad+\left(1+\beta_5^{-1}\right)\left(\mathcal{F}^T(x_k)-\zeta^T(\hat{x}_k)\hat{W}_k^T\right)\left(\mathcal{F}^T(x_k)\right.\\&\qquad\left.-\zeta^T(\hat{x}_k)\hat{W}_k^T\right)+\left(1+\beta_6^{-1}\right)L_k\left(\Re_k-\varpi_k\right)\check{C}x_k\\&\qquad\times x_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T+B\omega_k\omega_k^TB^T\\&\qquad+L_k\Re_k\check{D}\upsilon_k\upsilon_k^T\check{D}^T\Re_k^TL_k^T\Big\}. \qquad (22)\end{aligned}$$

Moreover, it can be readily observed that

$$\begin{aligned}\mathbb{E}\left\{x_kx_k^T\right\} &= \mathbb{E}\left\{(\tilde{x}_k+\hat{x}_k)\left(\tilde{x}_k^T+\hat{x}_k^T\right)\right\}\\&\le (1+\beta_6)P_k+\left(1+\beta_6^{-1}\right)\hat{x}_k\hat{x}_k^T. \qquad (23)\end{aligned}$$

Substituting (23) into (22) yields

$$\begin{aligned}&P_{k+1}\\&\le \mathbb{E}\Big\{\left(1+\beta_5+\beta_6\right)\left(\check{A}-L_k\varpi_k\check{C}\right)P_k\left(\check{A}^T-\check{C}^T\varpi_k^TL_k^T\right)\end{aligned}$$

$$\begin{aligned}\bar{\Phi}_{k+1} &= \bar{\Phi}_k\\&\quad+\theta_{k+1}^2\bigg(\mathrm{tr}\Big\{\varpi_{k+1}\check{A}\bar{\Phi}_k\check{A}^T\varpi_{k+1}^T\Big\}+\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}}+\Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\}\\&\qquad\qquad+\mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k\check{C}\bar{\Phi}_k\check{C}^T\varpi_k^TL_k^T\varpi_{k+1}^T\Big\}\\&\quad+\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\varpi_{k+1}L_k\Big(\Re_k-\varpi_k\Big)\bar{\Phi}_k\Big(\Re_k^T-\varpi_k^T\Big)L_k^T\varpi_{k+1}^T\\&\qquad+\Big(1+\beta_4^{-1}\Big)\varpi_{k+1}L_k\Big(\Re_k-\varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\Re_k^T-\varpi_k^T\Big)L_k^T\times\varpi_{k+1}^T\Big\}\\&\qquad+\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{A}\bar{\Phi}_k\check{A}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)+\Big(1+\beta_4^{-1}\Big)\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\\&\qquad\qquad\times\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\Big\}+\mathrm{tr}\Big\{\Big(\Re_{k+1}-\varpi_{k+1}\Big)\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\}\\&\qquad+2\varpi_{k+1}BQ_kB^T\varpi_{k+1}^T+2\varpi_{k+1}L_k\Re_k\check{D}G_k\check{D}^T\times\Re_k^TL_k^T\varpi_{k+1}^T+2\frac{\Re_{k+1}\check{D}G_{k+1}\check{D}^T\Re_{k+1}^T}{\left\|\check{C}^T\check{C}\right\|}\\&\qquad+2\Big(\Re_{k+1}-\varpi_{k+1}\Big)BQ_kB^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\bigg)\\&\quad+\theta_{k+1}\bigg(6\beta_1\bar{\Phi}_k+\Big(5\beta_2+\beta_1^{-1}\Big)\mathrm{tr}\Big\{\varpi_{k+1}\check{A}\bar{\Phi}_k\check{A}^T\varpi_{k+1}^T\Big\}\\&\qquad\quad+\Big(\beta_2+3\beta_3+\beta_1^{-1}+\beta_2^{-1}\Big)\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}}+\Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\}\\&\qquad+\Big(3\beta_3+\beta_1^{-1}+1\beta_2^{-1}\Big)\mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k\check{C}\bar{\Phi}_k\check{C}^T\varpi_k^TL_k^T\times\varpi_{k+1}^T\Big\}+\Big(2\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}\Big)\\&\qquad\times\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}L_k\Big(\Re_k-\varpi_k\Big)\bar{\Phi}_k\Big(\Re_k^T-\varpi_k^T\Big)L_k^T\varpi_{k+1}^T+\Big(1+\beta_4^{-1}\Big)\varpi_{k+1}L_k\Big(\Re_k-\varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\Re_k^T-\varpi_k^T\Big)L_k^T\\&\qquad\qquad\times\varpi_{k+1}^T\Big\}+\Big(\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+\beta_4^{-1}\Big)\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{A}\bar{\Phi}_k\check{A}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\\&\qquad\qquad\qquad+\Big(1+\beta_4^{-1}\Big)\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\Big\}\\&\qquad+\Big(\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+2\beta_4^{-1}\Big)\mathrm{tr}\Big\{\Big(\Re_{k+1}-\varpi_{k+1}\Big)\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\}\bigg). \qquad (16)\end{aligned}$$

$$
\begin{aligned}
&\times \left(1+\beta_5^{-1}+\beta_5\right)\bar{\mathcal{F}}I + \left(1+\beta_5^{-1}\right)^2 \hat{W}_k\zeta\left(\hat{x}_k\right)\zeta^T\left(\hat{x}_k\right)\hat{W}_k^T \\
&\times \left(1+\beta_6^{-1}\right)L_k\left(\Re_k-\varpi_k\right)\check{C}\left(\tilde{x}_k+\hat{x}_k\right)\left(\tilde{x}_k^T+\hat{x}_k^T\right)\check{C}^T\left(\Re_k^T\right. \\
&\left.-\varpi_k^T\right)L_k^T + B\omega_k\omega_k^T B^T + L_k\Re_k\check{D}\upsilon_k\upsilon_k^T\check{D}^T\Re_k^T L_k^T\} \\
\le&\left(1+\beta_5+\beta_6\right)\left(\check{A}-L_k\varpi_k\check{C}\right)P_k\left(\check{A}^T-\check{C}^T\varpi_k^T L_k^T\right) \\
&+\left(1+\beta_5^{-1}+\beta_5\right)\bar{\mathcal{F}}I + \left(1+\beta_5^{-1}\right)^2\hat{W}_k\zeta\left(\hat{x}_k\right)\zeta^T\left(\hat{x}_k\right)\hat{W}_k^T \\
&+\left(2+\beta_6^{-1}+\beta_6\right)L_k\left(\Re_k-\varpi_k\right)\check{C}P_k\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T \\
&+\left(1+\beta_6^{-1}\right)^2 L_k\left(\Re_k-\varpi_k\right)\check{C}\hat{x}_k\hat{x}_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right)L_k^T \\
&+BQ_kB^T + L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T L_k^T. \qquad (24)
\end{aligned}
$$

Finally, it follows from (24) that:

$$P_{k+1} \le \bar{P}_{k+1}$$

which ends the proof. ■

Now, we are in the position to design the gain matrix of the NN-based recursive estimator, which ensures that the UB obtained in Theorem 3 is minimized.

Theorem 4: The UB of the estimation error covariance can be minimized by adopting the following estimator gain parameter:

$$
\begin{aligned}
L_k =& \left(1+\beta_5+\beta_6\right)\check{A}\bar{P}_k\check{C}^T \\
&\times \Big(\left(1+\beta_5+\beta_6\right)\varpi_k\check{C}\bar{P}_k\check{C}^T \\
&\quad+\left(2+\beta_6^{-1}+\beta_6\right)\left(\Re_k-\varpi_k\right)\check{C}\bar{P}_k\check{C}^T\left(\Re_k^T-\varpi_k^T\right) \\
&\quad+\left(1+\beta_6^{-1}\right)^2\left(\Re_k-\varpi_k\right)\check{C}\hat{x}_k\hat{x}_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right) \\
&\quad+\Re_k\check{D}G_k\check{D}^T\Re_k^T\Big)^{-1}. \qquad (25)
\end{aligned}
$$

Proof: Taking the partial derivative of $\mathrm{tr}\{\bar{P}_{k+1}\}$ with respect to L_k and letting the derivative be zero, one has

$$
\begin{aligned}
&\frac{\partial\bar{P}_{k+1}}{\partial L_k} \\
&= -\left(1+\beta_5+\beta_6\right)\left(\check{A}-L_k\varpi_k\check{C}\right)\bar{P}_k\check{C}^T \\
&\quad+\left(2+\beta_6^{-1}+\beta_6\right)L_k\left(\Re_k-\varpi_k\right)\check{C}\bar{P}_k\check{C}^T\left(\Re_k^T-\varpi_k^T\right) \\
&\quad+\left(1+\beta_6^{-1}\right)^2 L_k\left(\Re_k-\varpi_k\right)\check{C}\hat{x}_k\hat{x}_k^T\check{C}^T\left(\Re_k^T-\varpi_k^T\right) \\
&\quad+L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T.
\end{aligned}
$$

Then, it is easy to see that $\bar{P}_{k+1}$ is minimized if the estimator gain L_k is selected as (25). ■

$$
\begin{aligned}
&\Phi_{k+1} \\
&= \mathrm{tr}\Big\{\mathbb{E}\Big\{\tilde{W}_{k+1}\tilde{W}_{k+1}^T\Big\}\Big\} \\
&\le \mathrm{tr}\Big\{\mathbb{E}\Big\{\Phi_k+\theta_{k+1}^2 \\
&\qquad\times\Big(\check{C}^T\varpi_{k+1}\check{C}\check{A}\tilde{x}_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\tilde{x}_k^T\check{A}^T\check{C} \\
&\qquad\times\varpi_{k+1}^T\check{C}+\check{C}^T\varpi_{k+1}\check{C}\Big(\mathcal{F}\left(x_k\right)-\hat{W}_k\zeta\left(\hat{x}_k\right)\Big)\tilde{\zeta}^T\left(\hat{x}_k\right) \\
&\qquad\times\tilde{\zeta}\left(\hat{x}_k\right)\Big(\mathcal{F}^T\left(x_k\right)-\zeta^T\left(\hat{x}_k^T\right)\hat{W}_k^T\Big)\check{C}^T\varpi_{k+1}^T\check{C} \\
&\qquad+\check{C}^T\varpi_{k+1}\check{C}L_k\varpi_k\check{C}\tilde{x}_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\tilde{x}_k^T\check{C}^T\varpi_k^T L_k^T \\
&\qquad\times\check{C}^T\varpi_{k+1}^T\check{C}+\check{C}^T\varpi_{k+1}\check{C}L_k\Big(\Re_k-\varpi_k\Big)\check{C}x_k\tilde{\zeta}^T\left(\hat{x}_k\right) \\
&\qquad\times\tilde{\zeta}\left(\hat{x}_k\right)x_k^T\check{C}^T\Big(\Re_k^T-\varpi_k^T\Big)L_k^T\check{C}^T\varpi_{k+1}^T\check{C} \\
&\qquad+\check{C}^T\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{C}\check{A}x_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)x_k^T\check{C}^T\check{A}^T\check{C}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\check{C} \\
&\qquad+\check{C}^T\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{C}\mathcal{F}\left(x_k\right)\tilde{\zeta}^T\left(\hat{x}_k\right)\times\tilde{\zeta}\left(\hat{x}_k\right)\mathcal{F}^T\left(x_k\right)\check{C}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\check{C}+2\varpi_{k+1}B \\
&\qquad\times Q_kB^T\varpi_{k+1}^T+2\varpi_{k+1}L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T L_k^T\varpi_{k+1}^T+2\frac{\Re_{k+1}\check{D}G_{k+1}\check{D}^T\Re_{k+1}^T}{\|\check{C}^T\check{C}\|}+2\Big(\Re_{k+1}-\varpi_{k+1}\Big)BQ_k \\
&\qquad\times B^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\Big)+\theta_{k+1}\Big(6\beta_1\tilde{W}_k\tilde{W}_k^T+\Big(5\beta_2+\beta_1^{-1}\Big)\check{C}^T\varpi_{k+1}\check{C}\check{A}\tilde{x}_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\tilde{x}_k^T\check{A}^T\check{C}\varpi_{k+1}^T\check{C} \\
&\qquad+\Big(\beta_2+3\beta_3+\beta_1^{-1}+\beta_2^{-1}\Big)\check{C}^T\varpi_{k+1}\check{C}\Big(\mathcal{F}\left(x_k\right)-\hat{W}_k\zeta\left(\hat{x}_k\right)\Big)\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\Big(\mathcal{F}^T\left(x_k\right)-\zeta^T\left(\hat{x}_k^T\right)\hat{W}_k^T\Big) \\
&\qquad\times\check{C}^T\varpi_{k+1}^T\check{C}+\Big(3\beta_3+\beta_1^{-1}+2\beta_2^{-1}\Big)\check{C}^T\varpi_{k+1}\times\check{C}L_k\varpi_k\check{C}\tilde{x}_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\tilde{x}_k^T\check{C}^T\varpi_k^T L_k^T\check{C}^T\varpi_{k+1}^T\check{C} \\
&\qquad+\Big(\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}\Big)\check{C}^T\varpi_{k+1}\check{C}L_k\times\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{C}\check{A}x_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)x_k^T\check{C}^T \\
&\qquad\times\Big(\Re_k^T\times-\varpi_k^T\Big)L_k^T\check{C}^T\varpi_{k+1}^T\check{C}+\Big(\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+\beta_4^{-1}\Big)\check{C}^T\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{C}\check{A}x_k\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right) \\
&\qquad\times x_k^T\check{C}^T\check{A}^T\check{C}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\check{C}+\Big(\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+2\beta_4^{-1}\Big)\check{C}^T\Big(\Re_{k+1}-\varpi_{k+1}\Big)\check{C}\mathcal{F}\left(x_k\right) \\
&\qquad\times\tilde{\zeta}^T\left(\hat{x}_k\right)\tilde{\zeta}\left(\hat{x}_k\right)\mathcal{F}^T\left(x_k\right)\check{C}^T\Big(\Re_{k+1}^T-\varpi_{k+1}^T\Big)\check{C}\Big)\Big\}\Big\}. \qquad (17)
\end{aligned}
$$

Based on the above discussion, the parameters, namely, L_k and θ_k, of the PNB state estimator for the CNs with unknown nonlinearities and EHMs are recursively computed by formulating and solving a minimization problem. The NN-based recursive state estimation algorithm can be summarized in Algorithm 1.

Remark 4: The main results of this article can be summarized in terms of the theorems as follows. In Theorem 1, a recursive NNW tuning parameter θ_k is designed to minimize the UB of the estimation error covariance Φ_k associated with the NNW, thereby enhancing the approximation capability for unknown nonlinearities. Theorem 3 establishes a recursive UB $\bar{P}_k$ for the estimation error covariance P_k of the system states, and Theorem 4 further derives the estimator gain matrix L_k by minimizing this UB $\bar{P}_k$. Together, these theorems constitute a unified recursive estimation framework, where both the estimator gain and the NNW tuning parameters are jointly updated in a recursive manner to improve estimation accuracy under the combined effects of unknown nonlinearities, EHMs, and partially accessible nodes.

Remark 5: A recursive state estimation algorithm has been developed in this article for CNs with unknown nonlinear dynamics and EHMs. The considered problem is novel. In comparison with existing research results, the distinctive contributions of this article are highlighted as follows.

1) While most existing research focuses on state estimation problems without considering the influence of communication schemes, the recursive state estimation problem for CNs with unknown nonlinear dynamics and EHMs is addressed in this article. This integration leads to a more practical model for real-world systems, in which such phenomena frequently occur.
2) NNs are utilized to approximate unknown nonlinear dynamics in CNs with partially accessible nodes. Unlike traditional approaches that rely on knowledge of nonlinear dynamics or linearization methods [31], [39], [53], the proposed approach employs NNs to address unknown nonlinear dynamics without requiring prior knowledge, offering enhanced flexibility and adaptability.
3) The major innovation of this article lies in the recursive updating of both the NNW tuning parameters and the estimator gain. A dynamic updating algorithm is embedded within a unified framework, which significantly improves the approximation accuracy and estimation performance in comparison with time-invariant or static methods presented in existing literature.
4) A novel NNW tuning law is provided for CNs with partially accessible nodes. Unlike prior research that often assumes measurement outputs from all sensor nodes are available [17], this research thoroughly investigates the impact of incomplete information on the approximation capabilities of NNs and develops a method to mitigate its negative effects.

IV. Illustrative Example

In this section, a simulation example is presented to verify the estimation performance under the presence of EHMs and unknown nonlinearities.

$$
\begin{aligned}
&\Phi_{k+1}\\
&\leq \mathbb{E}\Big\{\Phi_k + \theta_{k+1}^2\Big(\mathrm{tr}\Big\{\varpi_{k+1}\check{A}\Phi_k\check{A}^T\varpi_{k+1}^T\Big\}\\
&\qquad\qquad + \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}} + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\} + \mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k\check{C}\Phi_k\check{C}^T\varpi_k^TL_k^T\varpi_{k+1}^T\Big\}\\
&\quad + \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}L_k\Big(\mathfrak{R}_k-\varpi_k\Big)\Phi_k\Big(\mathfrak{R}_k^T-\varpi_k^T\Big)L_k^T\varpi_{k+1}^T + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}L_k\Big(\mathfrak{R}_k-\varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\mathfrak{R}_k^T-\varpi_k^T\Big)L_k^T\varpi_{k+1}^T\Big\}\\
&\quad + \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\check{A}\Phi_k\check{A}^T\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big) + \Big(1+\beta_4^{-1}\Big)\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\\
&\qquad \times\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\Big\} + \mathrm{tr}\Big\{\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\}\\
&\qquad + 2\varpi_{k+1}BQ_kB^T\varpi_{k+1}^T + 2\varpi_{k+1}L_k\mathfrak{R}_k\check{D}G_k\check{D}^T\mathfrak{R}_k^T\times L_k^T\varpi_{k+1}^T + 2\frac{\mathfrak{R}_{k+1}\check{D}G_{k+1}\check{D}^T\mathfrak{R}_{k+1}^T}{\big\|\check{C}^T\check{C}\big\|}\\
&\qquad + 2\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)BQ_kB^T\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\Big) + \theta_{k+1}\Big(6\beta_1\Phi_k + \Big(5\beta_2+\beta_1^{-1}\Big)\mathrm{tr}\Big\{\varpi_{k+1}\check{A}\Phi_k\check{A}^T\varpi_{k+1}^T\Big\}\\
&\qquad\quad + \Big(\beta_2+3\beta_3+\beta_1^{-1}+\beta_2^{-1}\Big)\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\times\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}} + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\}\\
&\qquad\quad + \Big(3\beta_3+\beta_1^{-1}+1\beta_2^{-1}\Big)\mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k\check{C}\Phi_k\check{C}^T\varpi_k^TL_k^T\times\varpi_{k+1}^T\Big\} + \Big(2\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}\Big)\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\\
&\qquad\quad \times\varpi_{k+1}L_k\Big(\mathfrak{R}_k-\varpi_k\Big)\Phi_k\Big(\mathfrak{R}_k^T-\varpi_k^T\Big)L_k^T\varpi_{k+1}^T + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}L_k\Big(\mathfrak{R}_k-\varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\mathfrak{R}_k^T-\varpi_k^T\Big)L_k^T\times\varpi_{k+1}^T\Big\}\\
&\qquad\quad + \Big(\beta_4+\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+\beta_4^{-1}\Big)\mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\check{A}\Phi_k\check{A}^T\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\\
&\qquad\qquad\qquad\qquad\qquad\qquad + \Big(1+\beta_4^{-1}\Big)\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\Big\}\\
&\qquad\quad + \Big(\beta_1^{-1}+\beta_2^{-1}+2\beta_3^{-1}+2\beta_4^{-1}\Big)\times\mathrm{tr}\Big\{\Big(\mathfrak{R}_{k+1}-\varpi_{k+1}\Big)\Big(\mathfrak{R}_{k+1}^T-\varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\}\Big)\Big\}
\end{aligned}
\tag{18}
$$

A CN with five nodes is considered, where only the measurements from two nodes are accessible (i.e., $S = 5$ and $s_0 = 2$). The weight adjacency matrix of the CN is selected as follows:

$$\Theta = \begin{bmatrix} -0.1 & 0.06 & 0.05 & 0.06 & 0.06 \\ 0.15 & -0.1 & 0.05 & 0.06 & 0.06 \\ 0.15 & 0.2 & -0.02 & 0.01 & 0.01 \\ 0.36 & 0.36 & -0.02 & 0.01 & 0.01 \\ 0.36 & 0.36 & -0.02 & 0.01 & 0.01 \end{bmatrix}.$$

The other parameters of the CNs are set as follows:

$$A_1 = A_2 = \begin{bmatrix} 0.72 & 0.4 & 0 \\ 0 & 1.025 & 0.1 \\ 0 & 0.3 & 0.2 \end{bmatrix}, \quad B_1 = B_2 = \begin{bmatrix} 0.2 \\ -0.1 \\ -1 \end{bmatrix}$$

$$A_3 = A_4 = A_5 = \begin{bmatrix} 0.7 & 0.4 & 0 \\ 0 & 0.8 & 0.1 \\ 0 & 0.3 & 0.1 \end{bmatrix}, \Lambda = \begin{bmatrix} 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

$$C_1 = C_2 = \begin{bmatrix} 0.45 & -0.2 & 0 \\ -0.88 & 0.2 & -0.2 \end{bmatrix}, \quad D_1 = D_2 = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix}$$

$$C_3 = C_4 = C_5 = \begin{bmatrix} 0.42 & -0.19 & 0 \\ -0.9 & 0.22 & -0 \end{bmatrix}$$

$$D_3 = D_4 = D_5 = \begin{bmatrix} 0.21 \\ 0.08 \end{bmatrix}, \quad B_3 = B_4 = B_5 = \begin{bmatrix} 0.19 \\ -0.09 \\ -0.99 \end{bmatrix}.$$

The nonlinear function $f(x_{i,k})(i \in \{1,2,3,4,5\})$ is chosen as follows:

$$f\left(x_{i,k}\right) = 4\begin{bmatrix} \sin\left(x_{i,k}\right) \\ \cos\left(x_{i,k}\right) \\ \cos\left(x_{i,k}\right) \end{bmatrix}.$$

Denote $x_{r,z,k}$ as the zth row of $x_{r,k}$ and $\hat{x}_{j,t,k}$ as the tth row of $x_{j,k}$. The covariances of the process noise ω_k and the measurement noise υ_k are $Q_k = 0.09$ and $G_k = 0.04$.

Set the initial energy level of the sensor node i in CNs as 2, and the maximum energy storage capacity is 3. The initial amount of energy harvested is 2, i.e.,

$$\aleph_{1,1} = \aleph_{2,1} = 2, H_1 = H_2 = 3, h_{1,1} = h_{2,1} = 2.$$

Moreover, we assume that the energy harvested in each instant is already known.

Denote $x_{i,r,k}$ as rth row of $x_{i,k}$. Set $\alpha_1 = 0.9$, $\alpha_2 = 0.8$, $\alpha_3 = 0.8$, and $\alpha_4 = 0.7$. The activation function $\vartheta(\hat{x}_{i,k})(i \in \{1,2,3,4,5\})$ of the NN and the initial values of the NNW estimates are chosen as follows:

$$\vartheta\left(\hat{x}_{i,k}\right) = \begin{bmatrix} 0.2\tanh\left(\hat{x}_{i,1,k}\right) \\ 0.2\tanh\left(\hat{x}_{i,2,k}\right) \\ 0.2\tanh\left(\hat{x}_{i,3,k}\right) \end{bmatrix}.$$

$$\begin{aligned}
\theta_{k+1} = -\frac{1}{2}\Big(& 6\beta_1\bar{\Phi}_k + \left(5\beta_2 + \beta_1^{-1}\right)\mathrm{tr}\left\{\varpi_{k+1}\check{A}\bar{\Phi}_k\check{A}^T\varpi_{k+1}^T\right\} + \left(\beta_2 + 3\beta_3 + \beta_1^{-1} + \beta_2^{-1}\right) \\
&\times \mathrm{tr}\left\{\left(1+\beta_4\right) \times \varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}} + \left(1+\beta_4^{-1}\right)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\right\} \\
&+ \left(3\beta_3 + \beta_1^{-1} + 1\beta_2^{-1}\right)\mathrm{tr}\left\{\varpi_{k+1}L_k\varpi_k\check{C}\bar{\Phi}_k\check{C}^T\varpi_k^T L_k^T \times \varpi_{k+1}^T\right\} \\
&+ \left(2\beta_4 + \beta_1^{-1} + \beta_2^{-1} + 2\beta_3^{-1}\right)\mathrm{tr}\Big\{\left(1+\beta_4\right) \times \varpi_{k+1}L_k\left(\Re_k - \varpi_k\right)\bar{\Phi}_k\left(\Re_k^T - \varpi_k^T\right)L_k^T\varpi_{k+1}^T \\
&\qquad + \left(1+\beta_4^{-1}\right)\varpi_{k+1}L_k\left(\Re_k - \varpi_k\right)\hat{x}_k\hat{x}_k^T\left(\Re_k^T - \varpi_k^T\right)L_k^T\varpi_{k+1}^T\Big\} \\
&+ \left(\beta_4 + \beta_1^{-1} + \beta_2^{-1} + 2\beta_3^{-1} + \beta_4^{-1}\right)\mathrm{tr}\Big\{\left(1+\beta_4\right) \times \left(\Re_{k+1} - \varpi_{k+1}\right)\check{A}\bar{\Phi}_k\check{A}^T\left(\Re_{k+1}^T - \varpi_{k+1}^T\right) \\
&\qquad + \left(1+\beta_4^{-1}\right)\left(\Re_{k+1} - \varpi_{k+1}\right)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\left(\Re_{k+1}^T - \varpi_{k+1}^T\right)\Big\} \\
&+ \left(\beta_1^{-1} + \beta_2^{-1} + 2\beta_3^{-1} + 2\beta_4^{-1}\right)\mathrm{tr}\left\{\left(\Re_{k+1} - \varpi_{k+1}\right) \times \left(\Re_{k+1}^T - \varpi_{k+1}^T\right)\bar{\mathcal{F}}\right\}\Big)\Big(\mathrm{tr}\left\{\varpi_{k+1}\check{A}\bar{\Phi}_k\check{A}^T\varpi_{k+1}^T\right\} \\
&+ \mathrm{tr}\left\{\left(1+\beta_4\right)\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}} + \left(1+\beta_4^{-1}\right)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\right\} \\
&+ \mathrm{tr}\left\{\varpi_{k+1}L_k\varpi_k\check{C}\bar{\Phi}_k\check{C}^T\varpi_k^T L_k^T\varpi_{k+1}^T\right\} + \mathrm{tr}\Big\{\left(1+\beta_4\right) \times \varpi_{k+1}L_k\left(\Re_k - \varpi_k\right)\bar{\Phi}_k\left(\Re_k^T - \varpi_k^T\right)L_k^T\varpi_{k+1}^T + \Big(1 \\
&+ \beta_4^{-1}\Big)\varpi_{k+1}L_k\left(\Re_k - \varpi_k\right)\hat{x}_k\hat{x}_k^T\left(\Re_k^T - \varpi_k^T\right)L_k^T\varpi_{k+1}^T\Big\} \\
&+ \mathrm{tr}\Big\{\left(1+\beta_4\right)\left(\Re_{k+1} - \varpi_{k+1}\right)\check{A}\bar{\Phi}_k\check{A}^T\left(\Re_{k+1}^T - \varpi_{k+1}^T\right) \\
&+ \left(1+\beta_4^{-1}\right)\left(\Re_{k+1} - \varpi_{k+1}\right)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\left(\Re_{k+1}^T - \varpi_{k+1}^T\right)\Big\} \\
&+ \mathrm{tr}\left\{\left(\Re_{k+1} - \varpi_{k+1}\right)\left(\Re_{k+1}^T - \varpi_{k+1}^T\right)\bar{\mathcal{F}}\right\} \\
&+ 2\varpi_{k+1}BQ_kB^T\varpi_{k+1}^T + 2\varpi_{k+1}L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T \\
&\times L_k^T\varpi_{k+1}^T + 2\frac{\Re_{k+1}\check{D}G_{k+1}\check{D}^T\Re_{k+1}^T}{\|\check{C}^T\check{C}\|} \\
&+ 2\left(\Re_{k+1} - \varpi_{k+1}\right)BQ_kB^T\left(\Re_{k+1}^T - \varpi_{k+1}^T\right)\Big)^{-1}.
\end{aligned} \tag{19}$$

The initial values are selected as follows:

$$\hat{W}_{1,1} = \begin{bmatrix} 1 & 1 & 0.95 \\ 0.22 & -0.41 & 0.1 \\ 0.1 & 0.9 & 0.1 \end{bmatrix}, \quad x_{1,1} = \begin{bmatrix} 1.2 \\ 2.4 \\ 1.2 \end{bmatrix}$$

$$\hat{W}_{2,1} = \begin{bmatrix} 1 & 1 & 0.95 \\ 0.22 & -0.41 & 0.1 \\ 0.1 & 0.9 & 0.1 \end{bmatrix}, \quad x_{2,1} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.8 \end{bmatrix}$$

$$\hat{W}_{3,1} = \begin{bmatrix} 1 & 1 & 0.95 \\ 0.22 & -0.41 & 0.1 \\ 0.1 & 0.9 & 0.1 \end{bmatrix}, \quad x_{3,1} = \begin{bmatrix} 1 \\ 0.4 \\ 0.6 \end{bmatrix}$$

$$\hat{W}_{4,1} = \begin{bmatrix} 1 & 1 & 0.95 \\ 0.22 & -0.41 & 0.1 \\ 0.1 & 0.9 & 0.1 \end{bmatrix}, \quad x_{4,1} = \begin{bmatrix} 0.66 \\ 0.4 \\ 0.8 \end{bmatrix}$$

$$\hat{W}_{5,1} = \begin{bmatrix} 1 & 1 & 0.95 \\ 0.22 & -0.41 & 0.1 \\ 0.1 & 0.9 & 0.1 \end{bmatrix}, \quad x_{5,1} = \begin{bmatrix} 0.66 \\ 0.4 \\ 0.8 \end{bmatrix}$$

$$\hat{x}_{1,1} = \begin{bmatrix} 0.4 \\ 1.2 \\ 0.2 \end{bmatrix}, \hat{x}_{2,1} = \begin{bmatrix} 0.6 \\ 0.8 \\ 0.8 \end{bmatrix}, \hat{x}_{3,1} = \begin{bmatrix} 0.55 \\ 1 \\ 0.3 \end{bmatrix}$$

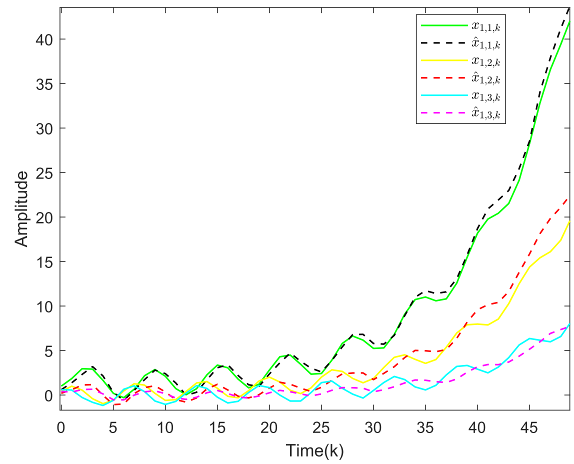

Fig. 2. State x_1 and its estimate.

$$\hat{x}_{4,1} = \begin{bmatrix} 0.53 \\ 1.05 \\ 0.27 \end{bmatrix}, \hat{x}_{5,1} = \begin{bmatrix} 0.53 \\ 1.05 \\ 0.27 \end{bmatrix}.$$

We aim to propose an NN-based recursive state estimation strategy to minimize the UB of $\mathbb{E}\{\tilde{x}_k \tilde{x}_k^T\}$ and $\mathrm{tr}\{\mathbb{E}\{\tilde{W}_k \tilde{W}_k^T\}\}$.

$$\begin{aligned}
\frac{\partial \bar{\Phi}_{k+1}}{\partial \theta_{k+1}} = 2\theta_{k+1}\Big(&\mathrm{tr}\Big\{\varpi_{k+1}\check{A}\bar{\Phi}_k\check{A}^T\varpi_{k+1}^T\Big\} \\
&+ \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\varpi_{k+1} \times \varpi_{k+1}^T\bar{\mathcal{F}} + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\} \\
&+ \mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k\check{C}\bar{\Phi}_k\check{C}^T\varpi_k^T L_k^T\varpi_{k+1}^T\Big\} \\
&+ \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\varpi_{k+1}L_k\Big(\Re_k - \varpi_k\Big)\bar{\Phi}_k\Big(\Re_k^T - \varpi_k^T\Big)L_k^T\varpi_{k+1}^T \\
&\qquad + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}L_k\Big(\Re_k - \varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\Re_k^T - \varpi_k^T\Big)L_k^T \times \varpi_{k+1}^T\Big\} \\
&+ \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\Re_{k+1} - \varpi_{k+1}\Big)\check{A}\bar{\Phi}_k\check{A}^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big) \\
&\qquad + \Big(1+\beta_4^{-1}\Big)\Big(\Re_{k+1} - \varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big)\Big\} \\
&+ \mathrm{tr}\Big\{\Big(\Re_{k+1} - \varpi_{k+1}\Big)\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\} \\
&+ 2\varpi_{k+1}BQ_kB^T\varpi_{k+1}^T + 2\varpi_{k+1}L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T \\
&\times L_k^T\varpi_{k+1}^T + 2\frac{\Re_{k+1}\check{D}G_{k+1}\check{D}^T\Re_{k+1}^T}{\|\check{C}^T\check{C}\|} + 2\Big(\Re_{k+1} - \varpi_{k+1}\Big)BQ_kB^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big)\Big) \\
&\times \mathrm{tr}\Big\{\varpi_{k+1}\check{A}\bar{\Phi}_k \times \check{A}^T\varpi_{k+1}^T\Big\} + \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\mathcal{F}} + \Big(1+\beta_4^{-1}\Big)\varpi_{k+1}\varpi_{k+1}^T\bar{\zeta}^2\hat{W}_k\hat{W}_k^T\Big\} \\
&+ \mathrm{tr}\Big\{\varpi_{k+1}L_k\varpi_k \times \check{C}\bar{\Phi}_k\check{C}^T\varpi_k^T L_k^T\varpi_{k+1}^T\Big\} + \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\varpi_{k+1}L_k\Big(\Re_k - \varpi_k\Big)\bar{\Phi}_k\Big(\Re_k^T - \varpi_k^T\Big)L_k^T\varpi_{k+1}^T \\
&+ \Big(1+\beta_4^{-1}\Big)\varpi_{k+1} \times L_k\Big(\Re_k - \varpi_k\Big)\hat{x}_k\hat{x}_k^T\Big(\Re_k^T - \varpi_k^T\Big)L_k^T\varpi_{k+1}^T\Big\} \\
&+ \mathrm{tr}\Big\{\Big(1+\beta_4\Big)\Big(\Re_{k+1} - \varpi_{k+1}\Big)\check{A}\bar{\Phi}_k\check{A}^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big) + \Big(1+\beta_4^{-1}\Big)\Big(\Re_{k+1} - \varpi_{k+1}\Big)\check{A}\hat{x}_k\hat{x}_k^T\check{A}^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big)\Big\} \\
&+ \mathrm{tr}\Big\{\Big(\Re_{k+1} - \varpi_{k+1}\Big)\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big)\bar{\mathcal{F}}\Big\} + 2\varpi_{k+1}B\omega_k \\
&\times \omega_k^T B^T\varpi_{k+1}^T + 2\varpi_{k+1}L_k\Re_k\check{D}G_k\check{D}^T\Re_k^T \\
&\times L_k^T\varpi_{k+1}^T + 2\frac{\Re_{k+1}\check{D}G_{k+1}\check{D}^T\Re_{k+1}^T}{\left\|\check{C}^T\check{C}\right\|} \\
&+ 2\Big(\Re_{k+1} - \varpi_{k+1}\Big)BQ_kB^T\Big(\Re_{k+1}^T - \varpi_{k+1}^T\Big).
\end{aligned}$$

Algorithm 1 NN-Based Recursive Estimation Algorithm

Step 1. Initialization: Set $k = 0$ and give the initial value y_0, $\hat{W}_0$, $\hat{x}_0$, $\bar{\Phi}_0$, $\bar{P}_0$.

Step 2. According to (25), compute the gain matrix L_k of the NN-based recursive estimator, based on which the UB of the estimation error covariance of system state $\bar{P}_{k+1}$ can be obtained.

Step 3. Based on $\hat{W}_k\zeta(\hat{x}_k)$ and L_k, compute the state estimate $\hat{x}_{k+1}$ by the recursion (8).

Step 4. Based on $\hat{x}_k$ and $\bar{\Phi}_k$, compute the NNW tuning parameter θ_{k+1} based on (19).

Step 5. Compute the NNW estimate $\hat{W}_{k+1}$ by the recursion (9). Then, the estimate of the unknown nonlinearities (i.e., $\hat{W}_{k+1}\zeta(\hat{x}_{k+1})$) can be calculated, and the UB $\bar{\Phi}_{k+1}$ can be obtained.

Step 6. Set $k = k + 1$ and go to *Step 2*

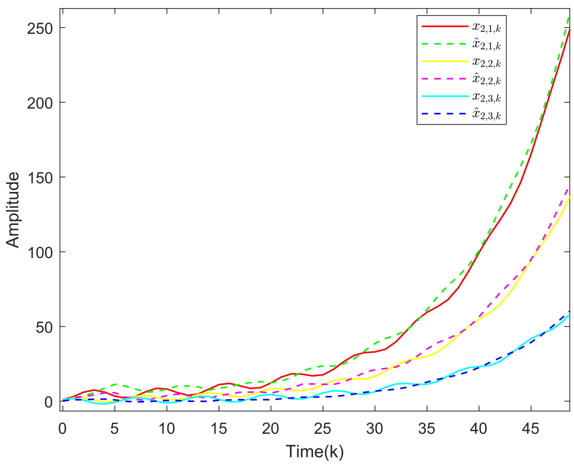

Fig. 3. State x_2 and its estimate.

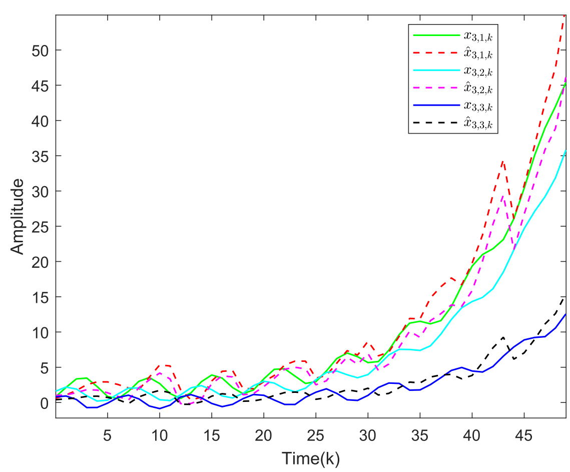

Fig. 4. State x_3 and its estimate.

Upon solving the estimation problem, the gain parameters and the tuning scalars of the NN-based estimator are recursively calculated. The simulation is conducted with a length of 50. The simulation results are presented in Figs. 2–15.

The trajectories of the system states and their estimates are illustrated in Figs. 2–6. It is observed that all nodes

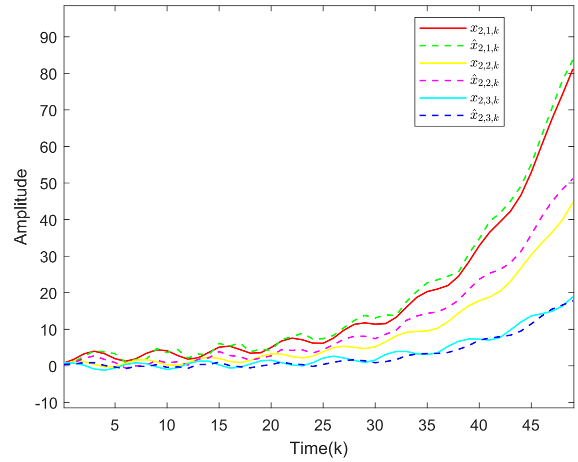

Fig. 5. State x_4 and its estimate.

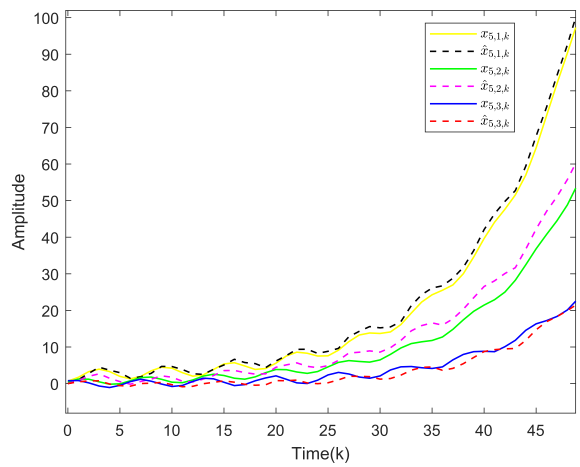

Fig. 6. State x_5 and its estimate.

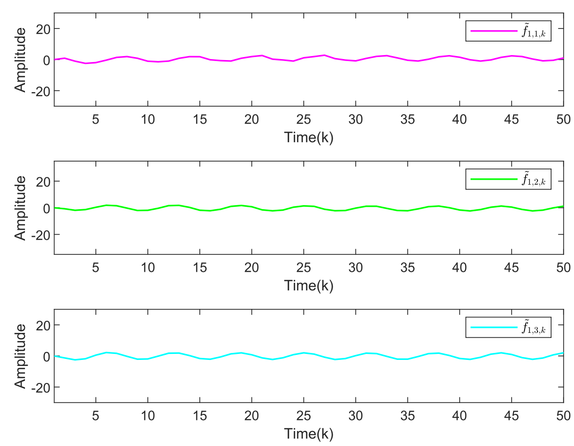

Fig. 7. Estimation error of the nonlinear function $f_1(x_k)$.

in the CN are unstable. Moreover, Figs. 7–11 depict the estimation errors of the unknown nonlinearities, defined as $\tilde{\mathcal{F}}(x_k) \triangleq \mathcal{F}(x_k) - \hat{W}_k\zeta(\hat{x}_k)$. It is evident that, regardless of whether the measurement information of the nodes is available, the NN-based recursive estimator demonstrates excellent estimation performance for both the system states and the unknown nonlinearities. Figs. 12 and 13 display the values

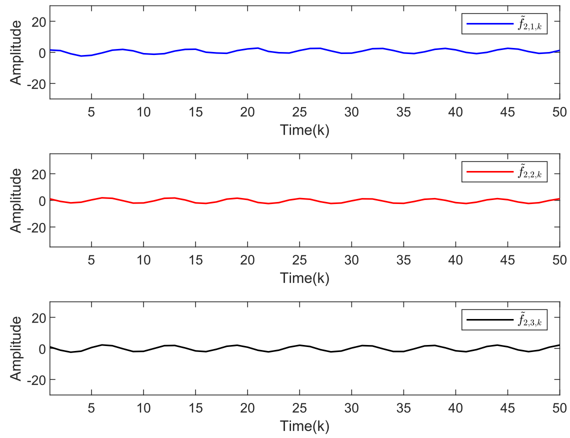

Fig. 8. Estimation error of the nonlinear function $f_2(x_k)$.

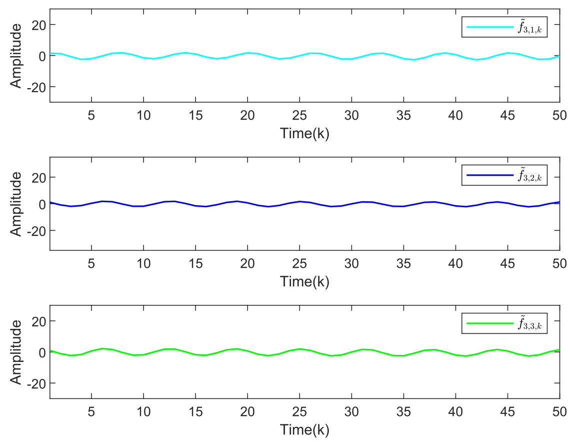

Fig. 9. Estimation error of the nonlinear function $f_3(x_k)$.

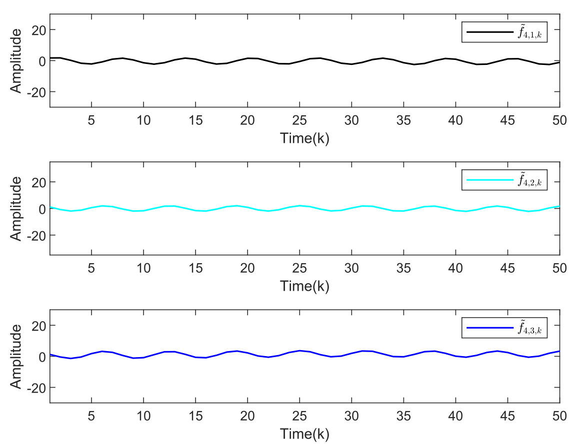

Fig. 10. Estimation error of the nonlinear function $f_4(x_k)$.

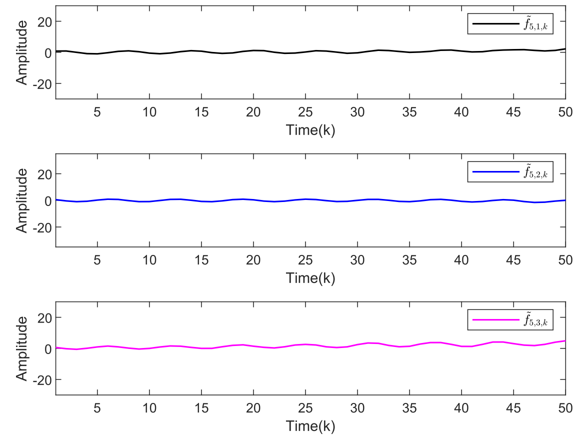

Fig. 11. Estimation error of the nonlinear function $f_5(x_k)$.

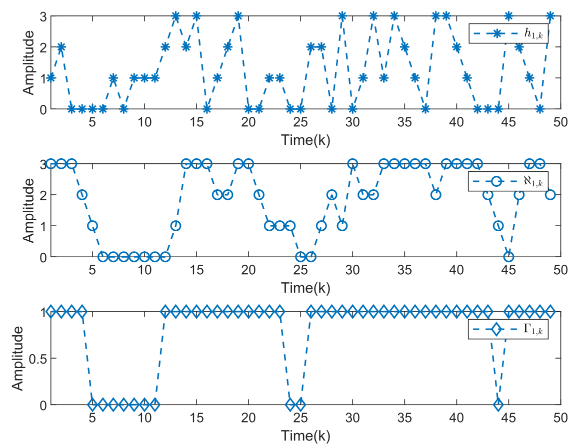

Fig. 12. Energy harvested and energy consumption of $x_{1,k}$.

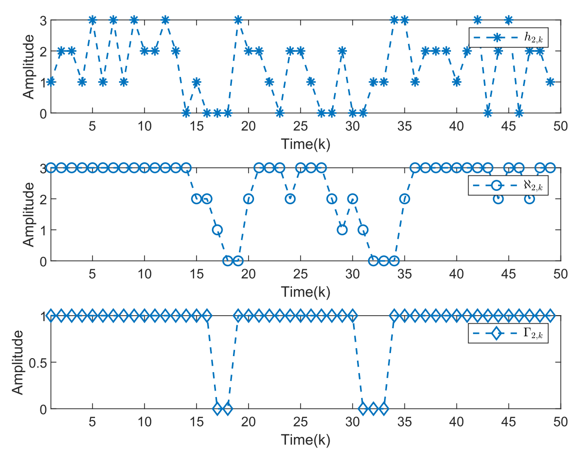

Fig. 13. Energy harvested and energy consumption of $x_{2,k}$.

of $\aleph_{i,k}$, random variable $h_{i,k}$, and $\Gamma_{i,k}$. In addition, to further examine how the main design parameters affect the estimation performance. Fig. 14 depicts the norm of the state estimation error when the NNW is set as a fixed value. It is obvious that fixed NNW makes the approximation performance worse, rather than the recursively calculated gain and NN tuning parameter maintain applicability to linear systems and systems with known nonlinear dynamics. Fig. 15 displays the norm of the state estimation error when considering different inner coupling matrices $\bar{\Lambda} = 0.6I$. It is easy to see that the developed state estimation strategy shows excellent estimation performance under various scenarios. All simulation results collectively confirm the effectiveness of the proposed estimation strategy in managing the challenges arising from the EHM and partially accessible nodes.

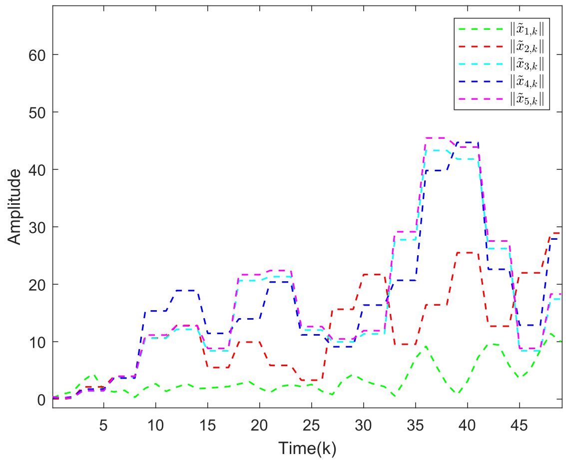

Fig. 14. Norm of the state estimation error when the NNW is fixed.

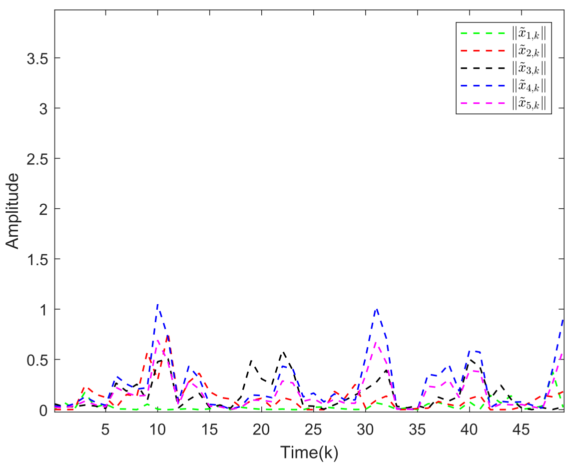

Fig. 15. Norm of the state estimation error when considering different inner coupling matrices $\bar{\Lambda}$.

V. Conclusion

This article has addressed the recursive state estimation problem for a class of energy harvesting CNs with partially accessible nodes and unknown nonlinear dynamics. To alleviate the challenges posed by energy constraints, a group of energy harvesting sensors has been utilized to collect energy from the external environment. The transmissions of measurement outputs from the energy harvesting sensors to the estimators have been conducted only when the energy level at the current instant has been sufficient to compensate for the energy consumption. Furthermore, NNs have been employed to approximate the unknown nonlinear dynamics of the CNs. An NN-based recursive state estimation strategy has been proposed to obtain the estimates of both the system states and the NNW. Within this strategy, the gain parameters and the NNW weight tuning scalars of the estimator have been recursively calculated in a unified framework. Sufficient conditions have been derived to ensure the minimization of the UBs of the estimation errors of the system states and the trace of the estimation errors of the NNWs. To verify the efficacy of the developed estimation strategy, a simulation example has been provided, which has demonstrated the applicability of the proposed approach. It should be pointed out that the results of this article can also be applied to the fault-tolerant state estimation problem subject to sensor failures. This is mainly due to the fact that sensor failures can be modeled by a sequence of binary variables that are similar to the sequence $\{\Re_{i,\aleph_{i,k}}\}_{k\geq 0}$. Future research will focus on the estimation problem for systems with EHMs and other phenomena such as cyberattacks, faults, packet loss, or time delays in networked systems [13], [18], [22], [28], [32], [40], [46], [50] and the pinning control problem for CNs with unknown nonlinearities [13], [17], where the essential effort will be devoted to the stability issues.

References

[1] A. Bounemeur and M. Chemachema, "Robust fuzzy adaptive fault-tolerant control for a class of second-order nonlinear systems," *Int. J. Adapt. Control Signal Process.*, vol. 39, no. 1, pp. 15–30, Jan. 2025.

[2] A. Bounemeur and M. Chemachema, "Adaptive fuzzy fault-tolerant control using Nussbaum-type function with state-dependent actuator failures," *Neural Comput. Appl.*, vol. 33, no. 1, pp. 191–208, Jan. 2021.

[3] A. Bounemeur, M. Chemachema, and N. Essounbouli, "Indirect adaptive fuzzy fault-tolerant tracking control for MIMO nonlinear systems with actuator and sensor failures," *ISA Trans.*, vol. 79, pp. 45–61, Aug. 2018.

[4] A. Bounemeur and M. Chemachema, "General fuzzy adaptive fault-tolerant control based on Nussbaum-type function with additive and multiplicative sensor and state-dependent actuator faults," *Fuzzy Sets Syst.*, vol. 468, Sep. 2023, Art. no. 108616.

[5] R. Caballero-Águila, J. Hu, and J. Linares-Pérez, "Filtering and smoothing estimation algorithms from uncertain nonlinear observations with time-correlated additive noise and random deception attacks," *Int. J. Syst. Sci.*, vol. 55, no. 10, pp. 2023–2035, Jul. 2024.

[6] S. Cai and J. Liang, "Recursive filtering for nonlinear systems with relay communication, energy harvesting and correlated noises," *Int. J. Netw. Dyn. Intell.*, vol. 4, no. 3, Sep. 2025, Art. no. 100021.

[7] L. Cao, W. Zhang, and H. Guo, "Finite-time cluster synchronisation of complex networks with multi-weights and nonlinear couplings via semi-intermittent control," *Int. J. Syst. Sci.*, vol. 56, no. 11, pp. 2624–2638, Aug. 2025.

[8] W. Chen, J. Hu, Z. Wu, and S. Ma, "A survey on fault detection for networked systems under communication constraints," *Syst. Sci. Control Eng.*, vol. 13, no. 1, Dec. 2025, Art. no. 2460434.

[9] G. Cybenko, "Approximation by superpositions of a sigmoidal function," *Math. Control Signals Syst.*, vol. 2, pp. 303–314, Jun. 1989.

[10] D. Dai, J. Li, J. Zhang, B. Jiang, and H. Dong, "Energy-harvesting recursive filtering for delayed systems with amplify-and-forward relays: Cooperative communication manner," *J. Franklin Inst.*, vol. 361, no. 18, Dec. 2024, Art. no. 107265.

[11] H. Fu, Z. Wang, D. Zhao, and B. Shen, "Proportional-integral state estimation for a class of nonlinear complex networks with uncertain couplings: Tackling adversarial replay attacks," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 5, pp. 4001–4012, Sep. 2025.

[12] H. Fu, Z. Wang, D. Zhao, and B. Shen, "Resilient state estimation for complex networks with partial node measurements and state saturations: Defense against replay attacks," *IEEE Internet Things J.*, vol. 12, no. 15, pp. 31881–31890, Aug. 2025.

[13] C. Gao, Z. Wang, L. Ma, H. Liu, and X. He, "Private synchronization control of complex networks against eavesdropping attacks: A differential privacy scheme," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 3, pp. 3028–3038, May 2024.

[14] P. Gao, C. Jia, and A. Zhou, "Encryption–decryption-based state estimation for nonlinear complex networks subject to coupled perturbation," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2357796.

[15] Y. Gao, J. Hu, H. Yu, C. Jia, and C. Chen, "Variance-constrained H_∞ state estimation for time-varying delayed neural networks with random access protocol and sensor failures," *Int. J. Netw. Dyn. Intell.*, vol. 4, no. 3, Sep. 2025, Art. no. 100019.

[16] Q. Ge, L. Zhang, Z. Zhao, X. Zhang, and Z. Lu, "Cubature particle filtering fusion with descent gradient and maximum correntropy for non-Gaussian noise," *Neurocomputing*, vol. 592, Aug. 2024, Art. no. 127634.

[17] Y. Guo, Z. Wang, J.-Y. Li, and Y. Xu, "Pinning synchronization for stochastic complex networks with randomly occurring nonlinearities: Tackling bit rate constraints and allocations," *IEEE Trans. Cybern.*, vol. 54, no. 12, pp. 7248–7259, Dec. 2024.

[18] A. Hasan, I. Kuncara, A. Widyotriatmo, O. Osen, and R. T. Bye, "Secure state estimation and control for autonomous ships under cyberattacks," *Syst. Sci. Control Eng.*, vol. 13, no. 1, Dec. 2025, Art. no. 2518964.

[19] K. Hornik, M. Stinchcombe, and H. White, "Multilayer feedforward networks are universal approximators," *Neural Netw.*, vol. 2, no. 5, pp. 359–366, Jan. 1989.

[20] N. Hou, H. Dong, W. Zhang, Y. Liu, and F. E. Alsaadi, "Event-triggered state estimation for time-delayed complex networks with gain variations based on partial nodes," *Int. J. Gen. Syst.*, vol. 47, no. 5, pp. 477–490, Jul. 2018.

[21] H. Kang, G. Zhai, and H. Nur Fadhilah, "Generalized decentralized control for robust stabilization and L_2 gain of nonlinear interconnected systems," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2313321.

[22] C. Li, Y. Liu, M. Gao, and L. Sheng, "Fault-tolerant formation consensus control for time-varying multi-agent systems with stochastic communication protocol," *Int. J. Netw. Dyn. Intell.*, Mar. 2024, Art. no. 100004.

[23] C. Li, Z. Wang, W. Song, S. Zhao, J. Wang, and J. Shan, "Resilient unscented Kalman filtering fusion with dynamic event-triggered scheme: Applications to multiple unmanned aerial vehicles," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 1, pp. 370–381, Jan. 2023.

[24] J. Li, X. Tian, and G. Wei, "Energy management for object tracking under the energy-harvesting: Hierarchical reinforcement learning method," *Appl. Math. Comput.*, vol. 479, Oct. 2024, Art. no. 128858.

[25] Q. Li, Z. Wang, N. Li, and W. Sheng, "A dynamic event-triggered approach to recursive filtering for complex networks with switching topologies subject to random sensor failures," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 10, pp. 4381–4388, Oct. 2020.

[26] N. Lin, D. Chen, J. Hu, C. Jia, and H. Yu, "Partial-nodes-based state estimation for state-saturated complex networks with random mixed couplings under dynamic event-triggered protocol," *Int. J. Adapt. Control Signal Process.*, vol. 37, no. 5, pp. 1203–1225, May 2023.

[27] D. Liu, Y. Liu, and F. E. Alsaadi, "Recursive state estimation based-on the outputs of partial nodes for discrete-time stochastic complex networks with switched topology," *J. Franklin Inst.*, vol. 355, no. 11, pp. 4686–4707, Jul. 2018.

[28] N. Liu and W. Qian, "Stability analysis of low-voltage direct current system with time-varying delay," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2293918.

[29] T.-J. Liu, Z. Wang, Y. Liu, and R. Wang, "Unscented-Kalman-filter-based remote state estimation for complex networks with quantized measurements and amplify-and-forward relays," *IEEE Trans. Cybern.*, vol. 54, no. 11, pp. 6819–6831, Nov. 2024.

[30] Y. Liu, Z. Wang, L. Ma, and F. E. Alsaadi, "A partial-nodes-based information fusion approach to state estimation for discrete-time delayed stochastic complex networks," *Inf. Fusion*, vol. 49, pp. 240–248, Sep. 2019.

[31] L. Ma, Z. Wang, Y. Liu, and F. E. Alsaadi, "Distributed filtering for nonlinear time-delay systems over sensor networks subject to multiplicative link noises and switching topology," *Int. J. Robust Nonlinear Control*, vol. 29, no. 10, pp. 2941–2959, Jul. 2019.

[32] C. Yang, "Security coordination control for the belt conveyor systems with false data injection attacks," *Int. J. Netw. Dyn. Intell.*, Mar. 2025, Art. no. 100001.

[33] X. Meng, Z. Wang, F. Wang, and Y. Chen, "Finite-horizon H_∞ state estimation for complex networks with uncertain couplings and packet losses: Handling amplify-and-forward relays," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17493–17503, Dec. 2024.

[34] W. Qian, D. Lu, S. Guo, and Y. Zhao, "Distributed state estimation for mixed delays system over sensor networks with multichannel random attacks and Markov switching topology," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8623–8637, Jun. 2024.

[35] W. Qian, Y. Wu, and J. Yang, "Event-driven reduced-order fault detection filter design for nonlinear systems with complex communication channel," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 1, pp. 281–293, Jan. 2024.

[36] F. Qu, E. Tian, and X. Zhao, "Chance-constrained H_∞ state estimation for recursive neural networks under deception attacks and energy constraints: The finite-horizon case," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 9, pp. 6492–6503, Sep. 2023.

[37] W. Song, Z. Wang, Z. Li, and H. Dong, "Multisensor particle filtering for nonlinear complex networks with heterogeneous measurements under non-Gaussian noises," *IEEE Trans. Cybern.*, vol. 56, no. 3, pp. 1241–1254, Mar. 2026, doi: 10.1109/TCYB.2025.3623631.

[38] W. Song, Z. Wang, Z. Li, Q.-L. Han, and D. Yue, "Maximum correntropy filtering for complex networks with uncertain dynamical bias: Enabling componentwise event-triggered transmission," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 17330–17343, Dec. 2024.

[39] W. Song, Z. Wang, Z. Li, J. Wang, and Q.-L. Han, "Nonlinear filtering with sample-based approximation under constrained communication: Progress, insights and trends," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 7, pp. 1539–1556, Jul. 2024.

[40] P. Sozhaeswari, R. Sakthivel, M. Chadli, and R. Abinandhitha, "Design of fault-tolerant non-fragile control for parabolic PDE systems with semi-Markov jump: The finite-time case," *Int. J. Syst. Sci.*, vol. 56, no. 11, pp. 2606–2623, Aug. 2025.

[41] Y.-C. Sun, K. Gao, L. Chen, F. Yang, and L. Yao, "Optimal power schedule for distributed Kalman filtering under DoS attacks: A Stackelberg game strategy," *Int. J. Syst. Sci.*, vol. 56, no. 9, pp. 2067–2081, Jul. 2025.

[42] Y. Theodor and U. Shaked, "Robust discrete-time minimum-variance filtering," *IEEE Trans. Signal Process.*, vol. 44, no. 2, pp. 181–189, Feb. 1996.

[43] E. Tian, Z. Wu, and X. Xie, "Codesign of FDI attacks detection, isolation, and mitigation for complex microgrid systems: An HBF-NN-based approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 5, pp. 6156–6165, May 2024.

[44] E. Tian, Y. Zou, and H. Chen, "Finite-time synchronization of complex networks with intermittent couplings and neutral-type delays," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 10, pp. 2026–2028, Oct. 2023.

[45] X. Wan, C. Yang, C.-K. Zhang, and M. Wu, "Hybrid adjusting variables-dependent event-based finite-time state estimation for two-time-scale Markov jump complex networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1487–1500, Feb. 2024.

[46] R. Wang and J. Liang, "The effect of multiscale parameters on the spiking properties of the morphological neuron with excitatory autapse," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2313865.

[47] X. Wang, D. Ding, X. Ge, and Q.-L. Han, "Supplementary control for quantized discrete-time nonlinear systems under goal representation heuristic dynamic programming," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3202–3214, Mar. 2024.

[48] B. Wei, E. Tian, Z. Gu, J. Zhai, and D. Liang, "Quasi-consensus control for stochastic multiagent systems: When energy harvesting constraints meet multimodal FDI attacks," *IEEE Trans. Cybern.*, vol. 54, no. 3, pp. 1755–1767, Mar. 2024.

[49] T. Xue and Y. Liu, "State estimation based on measurement from a part of nodes for a delayed Itô type of complex network with Markovian mode-dependent parameters," *Neurocomputing*, vol. 600, Oct. 2024, Art. no. 128188.

[50] H. Yan, K. Ding, Y. Zhao, X. Wu, and T. Huang, "Extended dissipative sampling control for singular systems with time-varying delay," *Int. J. Syst. Sci.*, vol. 56, no. 12, pp. 2978–2990, Sep. 2025.

[51] F. Yang, H. Dong, Y. Shen, X. Li, and D. Dai, "Protocol-based non-fragile state estimation for delayed recurrent neural networks subject to replay attacks," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 1, pp. 249–251, Jan. 2024.

[52] H. Yu, Z. Wang, L. Zou, and Y. Wang, "A survey on security control and estimation for cyber-physical systems under cyber-attacks: Advances, challenges and future directions," *Artif. Intell. Sci. Eng.*, pp. 1–16, 2025.

[53] S. Yuan, L. Ma, and C. Gao, "Probability-guaranteed consensus control for nonlinear multi-agent systems under bit flips," *Int. J. Netw. Dyn. Intell.*, vol. 4, no. 3, Sep. 2025, Art. no. 100020.

[54] J. Zhang, J. Lu, J. Liang, and K. Shi, "Privacy-preserving average consensus in multiagent systems via partial information transmission," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 53, no. 5, pp. 2781–2791, May 2023.

[55] R. Zhang, H. Liu, Y. Liu, and H. Tan, "Dynamic event-triggered state estimation for discrete-time delayed switched neural networks with constrained bit rate," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2334304.

[56] Y. Zhang, G. Liu, and X. Song, "Unscented recursive three-step filter based unbiased minimum-variance estimation for a class of nonlinear systems," *Int. J. Syst. Sci.*, vol. 56, no. 2, pp. 227–236, Jan. 2025.

[57] Y. Zhang, Z. Wang, L. Zou, W. Qian, and S. Du, "Neural-network-based recursive state estimation for nonlinear networked systems with binary-encoding mechanisms," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 6, pp. 10072–10083, Jun. 2025.

[58] D. Zhao, C. Gao, J. Li, H. Fu, and D. Ding, "PID control and PI state estimation for complex networked systems: A survey," *Int. J. Syst. Sci.*, vol. 56, no. 11, pp. 2735–2750, Aug. 2025.

[59] L. Zhao and B. Li, "Adaptive fixed-time control for multiple switched coupled neural networks," *Int. J. Netw. Dyn. Intell.*, vol. 2024, Sep. 2024, Art. no. 100018.

[60] L. Zou, Z. Wang, J. Hu, and H. Dong, "Partial-node-based state estimation for delayed complex networks under intermittent measurement outliers: A multiple-order-holder approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7181–7195, Oct. 2023.

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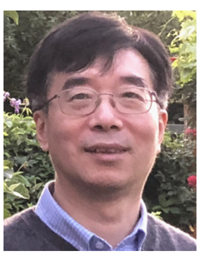

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